Cross-city propagation-model transfer for network planning in 6G digital twin networks: tile parameters versus site-aware link models

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Abstract

Network digital twins are expected to become a standard component of the planning and operation of sixth-generation (6G) mobile networks, and they require a propagation model for areas where no measurement campaign exists yet. Tile-based transfer fits a log-distance exponent and intercept per map tile and carries them to unmeasured tiles by descriptor similarity. This article evaluates such transfer across cities, and measures what its errors cost a network planner, on open ray-traced corpora: 11 geo-referenced cities with 220 rooftop sites at 3.5 GHz and in the upper mid-band at 7.5 GHz (8,977,400 and 7,038,989 links), replicates with other sitings and random building materials, and independently modelled external scenes. The tile labels are shown to be imprecise, with a cluster-robust uncertainty 3.2 times the naive one, and to change with the serving sites, the tile size and the dynamic range of the data. At the network level, even exact tile labels leave a path-loss error of 11.6 dB and assign the correct serving site to 50.1% of the locations. The gap is decomposed: a reachability model for distant sites removes most of the planning loss, a line-of-sight dual-slope label part of the remainder, and the rest is the geometry of each link. A site-aware hurdle model, which predicts every link and whether it exists from its obstruction and diffraction geometry, reaches 9.3 dB, 68.4% correct serving sites and a planning regret of 0.7% for a five-site plan in every held-out city, and outperforms per-link physics baselines in path loss and serving site under the same protocol. On scenes with other materials, a survey of 25 random tiles calibrates it to the accuracy of exact tile labels, and a distribution-free city-level certificate allows 80.6% of the locations to be promised as covered at a guaranteed risk, against 48.4% for tile transfer. All corpora, code and results are released, and every number, table and figure of the article is regenerated by one command.

Keywords: 6G, digital twin network, network planning, mobile networks, transfer learning, network simulation, coverage certification

1. Introduction

The IMT-2030 framework for sixth-generation (6G) mobile networks makes the network digital twin a first-class component of network planning and operation: a virtual replica of the network and its radio environment in which coverage, interference, cell layouts and site positions are evaluated before anything is built or measured [1, 2, 3]. The ITU-T architecture for digital twin networks places the models that feed such decisions in the twin's data and model layers [4]. For a radio access network, the central model is the one that maps a candidate site layout to the quantities a planner decides on: the serving site of every location (cell association), the signal-to-interference-plus-noise ratio (SINR), the covered share of the area, the position of the cell edges, which drives handover, and the number and positions of sites. This model is needed for areas where no measurement campaign exists yet, and for the new bands under study for 6G, the upper mid-band between 7 and 24 GHz [5], where campaigns are scarce. Three families of propagation models compete for this role. Deterministic ray tracing resolves the geometry but needs detailed three-dimensional data and must be rerun for every change of layout [6, 7]. Empirical formulas such as Okumura–Hata or 3GPP TR 38.901 are cheap but apply one parameter set to a whole environment class [8, 9]. Tile-based models sit between the two: the twin is partitioned into tiles, each tile carries the two parameters (γ, P_{L_0}) of a log-distance law, and parameters learned where measurements exist are transferred to unmeasured tiles by environmental similarity [10, 11]. The transported quantity is small and physically interpretable, and a planning tool can evaluate it for any site layout without further simulation.

Whether such transfer works *across cities*, and what its errors cost a network planner, has not been established. The descriptor-based framework of [11], which transfers tile parameters with a Gaussian kernel over a descriptor space, was evalu-

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ated on confidential measurements from two cities, one serving as the source of the other. Three questions remain open. First, how reliable are the tile labels themselves, given that a tile of 100 m sees its serving sites over a narrow range of distances and that neighbouring receivers share propagation paths? Second, how does kernel transfer behave when the target city differs from every source city, which is exactly when transfer is needed, and how does it compare with learned regressors under one protocol? Third, what do parameter errors mean for the quantities a planner uses, namely path loss, serving site, SINR, coverage, cell edges and the choice of sites; which part of the error is due to the tile representation and which part to the absence of a reachability model or of a line-of-sight state; and how much coverage can an operator promise from a transferred model?

To address these questions, this article presents a reproducible, network-level evaluation of cross-city propagation-model transfer for 6G digital twin networks on open ray-traced corpora. The main contributions of this article are the following.

- **Reproducible multi-city, multi-band corpora.** Eleven geo-referenced city scenes built from a pinned release of open building data are ray traced from 220 rooftop sites at 3.5 GHz (8,977,400 links, 10,428 labelled tiles) and at 7.5 GHz in the upper mid-band, with a multi-site ray-budget study that states the solver noise floor, a physical consistency check of the solver output, replicates with random and with mid-rise rooftop siting and with random building materials, and an external test on independently modelled scenes. Every number in this article is regenerated from the released data by one command, whose cost is stated.

- **Tile labels with their precision.** The per-tile fit is shown to be ill-conditioned; ad-hoc regularisation is replaced by empirical-Bayes shrinkage whose risk is bounded whatever the tile geometry (Proposition 1), with a small-sample (CR2) cluster-robust variance; and the labels are shown to depend on the serving sites, the tile size and the dynamic range of the data.

- **A leave-one-city-out benchmark with nested selection and city-level inference.** Standard-model, label-averaging, Gaussian-process, linear, tree, neural and domain-adaptation operators are compared under one protocol in which every hyper-parameter is chosen inside the training cities, with five seeds, with and without surveyed tiles, and every claim is tested at the level of cities with bootstrap intervals, Wilcoxon tests and Holm correction. An analysis of the kernel operator (Proposition 2) locates its weakness in the isotropic metric.

- **A decomposition of the network-level gap.** Every operator is evaluated through path loss, serving site, SINR, coverage, cell edges, dimensioning and site planning scored on the ray-traced truth over the complete receiver grid, outage locations included, and the contributions of the reachability cut-off, of the line-of-sight state and of the link geometry are separated by giving the tile model a hurdle and a dual-slope label.

- **Site-aware link transfer against physics baselines.** A hurdle link model conditioned on the obstruction geometry of each link outperforms the tile model with perfect labels in path loss, serving site and planning in every held-out city, and the non-learned per-link baselines (free space with single and multiple knife edges, 3GPP UMa with geometric line of sight) in path loss and serving site, across bands, sitings, materials and independently modelled scenes; a small unbiased survey calibrates its per-site offsets with an empirical-Bayes prior.

- **Coverage certificates.** A distribution-free, city-level coverage certificate from leave-one-city-out residuals (Proposition 4) states what each transfer model can promise at a stated risk.

The remainder of this article is organised as follows. Section 2 reviews the related work and positions this study. Section 3 formalises the network model and the transfer problem. Section 4 describes the corpora, the tile labels and descriptors, the transfer operators, the site-aware link model and the evaluation methodology. Section 5 reports the experimental setting and the results, Section 6 discusses their implications and limitations, and Section 7 concludes the article and outlines future work.

2. Related work

The problem studied here lies at the intersection of three bodies of work: propagation models and the digital twins that consume them for network planning, learned and transferred propagation models, and distribution-free uncertainty for decisions taken on predicted quantities. This section reviews each in turn and closes with a feature-based comparison between the representative studies and the present work.

2.1. Propagation models and digital twins for network planning

Empirical models such as Okumura–Hata [8] and the 3GPP TR 38.901 formulas [9] express path loss through a small number of parameters fitted to measurement campaigns and applied uniformly over an environment class; TR 38.901 distinguishes a line-of-sight and a non-line-of-sight law, which a digital twin can select per link from its geometry. Measurement studies showed early that the path-loss exponent is itself a random quantity that changes from cell to cell [12], which is the variability a tile-level model tries to capture. Diffraction over the buildings along a link is described by the knife-edge and multiple-edge constructions of ITU-R P.526 [13], which serve in this study as non-learned per-link baselines. Deterministic ray tracing resolves the geometry explicitly [6]; differentiable ray tracers such as Sionna RT make city-scale simulation and calibration against measurements practical [7, 14]. In this study,

ray tracing produces the ground truth for many cities and is not the run-time predictor.

Digital twins of wireless networks [2, 3] and the ITU-T digital twin network architecture [4] need propagation models for regions where no measurements exist yet and consume them through planning functions. Site selection with propagation predictions is a classical optimisation problem [15]; coverage maximisation is submodular and admits greedy approximation guarantees [16]. In operational networks, targeted measurements can be collected with the minimisation-of-drive-tests (MDT) framework [17, 18], which motivates the few-shot setting of this study; choosing where to measure is a submodular sensor-placement problem [19].

2.2. Learned and transferred propagation models

Convolutional [20], autoencoder [21], adversarial [22] and diffusion [23] models predict dense radio maps from environment maps, and radio-map estimation from sparse measurements has a mature statistical treatment [24]. Channel knowledge maps [25] generalise the idea to other channel quantities. Regression of path loss from engineered or image features has been studied with measurements at 2.6 GHz [26] and 28 GHz [27], across several cities with obstruction-height maps [28], and is surveyed in [29]. These methods predict path loss or maps directly; the tile-parameter approach studied here predicts a compact, physically interpretable pair per tile that a planning tool can evaluate for any site layout. The link model of this study belongs to the feature-regression family of [26, 28]; its purpose is to serve as a strong, simple and interpretable reference against which the tile representation is measured.

MAPLE divides 3-D city maps into tiles and estimates mixed-path propagation in a new deployment area from the similarity of its tiles to tiles of measured areas [10]. The descriptor-based framework of [11] describes each tile by 116 terrain, built-structure, land-cover and climatic descriptors, calibrates log-distance parameters per tile from measurements with the mixed-path composition of MAPLE, and transfers them to unmeasured tiles by a Gaussian-kernel weighting of descriptor similarity. It was evaluated with confidential measurements of an industrial partner in two European cities, each serving once as the source of the other, with tiles of 200 m to 1000 m, by the normalised error of the received power, and against empirical models only. Transfer between domains with different input distributions is covariate shift [30], and its difficulty depends on the discrepancy between the domains [31]; importance weighting [30] and the alignment of feature statistics [32] are standard remedies, which are included here as baselines.

2.3. Uncertainty, censoring and positioning of this study

Conformal prediction gives distribution-free prediction sets under exchangeability [33], and leave-one-out variants such as the jackknife+ reuse every training point for calibration [34]; this study applies the exchangeability argument at the level of cities, the unit of shift. Measurements with a detection floor are censored, for which the Tobit model is the classical treatment [35]; received-power data beyond a receiver's sensitivity are censored in exactly this sense, and a hurdle model that first predicts whether a link exists is the natural counterpart for prediction.

Table 1 summarises how the present study relates to the representative works above. None of the learned propagation models has been scored through the quantities a planner decides on, none states what coverage can be promised from a transferred model, and the tile-transfer framework has been evaluated on two cities with confidential data. The present study addresses the same transfer problem on a larger public corpus with a re-implementation of the kernel operator on the built-structure descriptors, analyses that operator, compares it with learned regressors under a common protocol, evaluates both the receiver-tile and the mixed-path composition, measures the consequences for network planning, and states the coverage that can be promised at a given risk.

3. System model and problem formulation

This section fixes the network model through which every predictor is scored, defines the tile-parameter representation and the transfer problem, and derives the risk decomposition that governs label-averaging transfer operators. Table 2 summarises the notation.

3.1. Network model

A city contains a set $\mathcal{B}$ of base-station sites and a set $\mathcal{P}$ of receiver locations (pixels) at user height. The link (p, b) has three-dimensional length d_{pb} and path loss $P_L(p, b)$ in dB. With transmit power P_{tx} (dBm) at every site and noise power P_{N}, the received power, the serving site and the downlink signal-to-interference-plus-noise ratio under full load are

$$P_{pb} = P_{\mathrm{tx}} - P_L(p, b), \qquad s(p) = \arg\max_{b \in \mathcal{B}} P_{pb},$$
$$\mathrm{SINR}(p) = \frac{10^{P_{p\,s(p)}/10}}{\sum_{b \neq s(p)} 10^{P_{pb}/10} + 10^{P_{\mathrm{N}}/10}}. \tag{1}$$

A link whose path loss exceeds the coupling-loss cut-off $L_{\max}$ of the receiver is not received at all; a pixel that no site reaches is an outage pixel, with $s(p)$ undefined and its SINR set to the outage floor. A pixel is *covered* at coupling-loss level L if its serving path loss is at most L, and the covered share of the city is the mean of the coverage indicator,

$$c_L(p) = \mathbb{1}\{P_L(p, s(p)) \leq L\}, \qquad C_L = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} c_L(p). \tag{2}$$

The spectral efficiency of a pixel follows the attenuated and truncated Shannon mapping used in 3GPP system evaluations [36],

$$\eta = \begin{cases} 0, & \mathrm{SINR} < -10\,\mathrm{dB}, \\ \min\{0.6 \log_2(1 + \mathrm{SINR}),\ 4.4\}, & \text{else}, \end{cases} \tag{3}$$

in b/s/Hz, evaluated at every pixel with the SINR in linear units inside the logarithm. Mobility decisions depend on where

Table 1: Feature-based comparison between representative related studies and the present work. Cross-city: trained on some cities and evaluated on others. Tile parameters: a per-tile log-distance pair that a planning tool can reuse. Link geometry: per-link obstruction or image features. Network level: serving site, SINR, coverage or cell edges scored against the truth. Planning: site selection or dimensioning. Guarantee: a statistical coverage guarantee for the transferred model.

Reference	Cross-city	Tile parameters	Link geometry	Network level	Planning	Guarantee	Main limitation with respect to this study
MAPLE [10]	✓	✓	–	–	–	–	Tile-similarity estimation of mixed-path propagation; evaluated through the propagation estimate rather than through planning quantities.
Ozyurt [11]	✓	✓	–	–	–	–	Two cities with confidential measurements, each the source of the other; received-power error against empirical models only.
Thrane et al. [26]	–	–	✓	–	–	–	Measurements from one area at 2.6 GHz; path loss only.
Gupta et al. [27]	–	–	✓	–	–	–	One city at 28 GHz; path loss only.
Dempsey et al. [28]	✓	–	✓	–	–	–	Path loss only; no reusable parameter representation and no planning quantities.
RadioUNet [20]	–	–	–	–	–	–	Dense map prediction on simulated maps, split at random rather than held out by city.
RadioDiff [23]	–	–	–	–	–	–	Generative map prediction on simulated maps; no network-level evaluation.
Hoydis et al. [14]	–	–	✓	–	–	–	Calibration of a ray tracer to measurements of one area; the ray tracer remains the predictor.
Amaldi et al. [15]	–	–	–	✓	✓	–	Optimisation with given propagation predictions; no learned or transferred model.
This work	✓	✓	✓	✓	✓	✓	Ray-traced rather than measured truth; eleven cities limit the power of city-level tests.

Table 2: Notation.

Symbol	Description
$\mathcal{B}, \mathcal{P}$	Sites of a city and receiver locations (pixels)
d_{pb}, $P_L(p,b)$	3-D length and path loss (dB) of link (p,b)
P_{tx}, P_{N}	Transmit power and noise power (dBm)
P_{pb}, $s(p)$	Received power from site b and serving site of pixel p
$L_{\max}$, L	Coupling-loss cut-off of the receiver and coverage level
$c_L(p)$, C_L	Coverage indicator of pixel p and covered share at level L
$\eta(p)$, $\mathcal{E}$	Spectral efficiency of pixel p and set of cell-edge pixels
G_s, t	Tile side and tile index
γ_t, $P_{L_0,t}$, d_0	Exponent, intercept and reference distance of tile t
$\boldsymbol{\theta}_t$, $\hat{\boldsymbol{\theta}}_t$	Tile label $(\gamma_t, P_{L_0,t})^\top$ and its estimate
b_t, v_t	Least-squares slope of tile t and its sampling variance
w_t, μ, τ^2	Shrinkage weight; prior mean and variance of the slopes
$\mathbf{v}_t$	Descriptor of tile t (p features)
$\mathcal{T}_{\mathrm{S}}$, N	Source tiles and their number
$\mathcal{A}$, $w_j(\mathbf{v})$	Transfer operator and label-averaging weights
n_{eff}, σ	Effective neighbourhood size and kernel bandwidth
k	Number of surveyed target tiles
κ, o_s	Prior strength and offset of site s in the link calibration
K, R_K	Number of sites selected by the planner and planning regret
α, $\hat{q}_t$	Risk level and margin of the coverage certificate

the serving site changes: a pixel belongs to the set $\mathcal{E}$ of cell-edge pixels when its eastern or northern neighbour has a different serving site, so that every boundary crossing is counted once, and $|\mathcal{E}|$ is proportional to the number of handovers a route through the city would trigger. Every quantity in Eqs. (1)–(3) and the set $\mathcal{E}$ are evaluated for the ray-traced truth and for every predicted path-loss map $\widehat{P_L}(p,b)$; the comparison of the two is the network-level evaluation of Section 4.6.

3.2. Tile-parameter model and the transfer problem

The city is partitioned into square tiles of side G_s. Inside tile t the path loss follows the log-distance law [37, 12]

$$P_L(p,b) = P_{L_0,t} + 10\,\gamma_t \log_{10}\frac{d_{pb}}{d_0} + \varepsilon_{pb}, \qquad p \in t, \tag{4}$$

with exponent γ_t, intercept $P_{L_0,t}$ at the reference distance d_0 and a residual ε_{pb} of variance σ^2_ε. The reference distance is set to $d_0 = 100\,\mathrm{m}$, the tile scale, so that $P_{L_0,t}$ is an interpolated rather than an extrapolated quantity; the intercept at 1 m follows as $P_{L_0,t} - 20\gamma_t$. Writing $x = 10\log_{10}(d/d_0)$, Eq. (4) is linear in $\boldsymbol{\theta}_t = (\gamma_t, P_{L_0,t})^\top$ with design matrix $\mathbf{A}_t = [\mathbf{x}\ \mathbf{1}]$.

Equation (4) is the interface between the digital twin and network planning: a planning tool that knows $\boldsymbol{\theta}_t$ for every tile and the site positions can evaluate Eq. (1) anywhere without ray tracing. Two structural consequences of the receiver-tile model are used later. With the parameters of the receiver's tile applied to every site and $\gamma_t > 0$, received power decreases with distance, so (i) the predicted serving site is the nearest site, $s(p) = \arg\min_b d_{pb}$, whatever $\boldsymbol{\theta}_t$, and (ii) the predicted SINR is

$$\mathrm{SINR}(p) = \frac{(d_{ps}/d_0)^{-\gamma_t}}{\sum_{b\neq s}(d_{pb}/d_0)^{-\gamma_t} + 10^{(P_{\mathrm{N}}-P_{\mathrm{tx}}+P_{L_0,t})/10}}, \tag{5}$$

which in the interference-limited regime depends on the exponent only, while coverage at level L depends on both parameters. Property (i) bounds what any receiver-tile predictor can achieve in cell association: the metric is identical for every transfer operator and measures the tile model itself. A receiver-tile model also has no notion of a link that is not received: every site reaches every pixel with a finite loss. Both limitations can be relaxed without leaving the tile representation, by a *hurdle* that removes the links a classifier predicts beyond $L_{\max}$ and by

a *dual-slope* label with one pair for line-of-sight and one for obstructed links; Section 5.5 measures what each buys. Mixed-path compositions [10, 11], which combine the parameters of every tile a link crosses, are not bound by property (i) and are evaluated in Section 5.6.

Each tile carries a descriptor $\mathbf{v}_t \in \mathbb{R}^p$ computed from the geometry of the digital twin. Measurements, and hence labels $\boldsymbol{\theta}_t$, are available in a set of *source* cities $\mathcal{T}_S = \{(\mathbf{v}_j, \boldsymbol{\theta}_j)\}_{j=1}^N$ and not in the *target* city, whose tiles have descriptors only. A *transfer operator* maps the source data and a target descriptor to a parameter estimate,

$$\hat{\boldsymbol{\theta}}_u = \mathcal{A}(\mathcal{T}_S; \mathbf{v}_u), \qquad u \in \text{target}, \tag{6}$$

and its target risk is $R = \mathbb{E}_{\mathbf{v}} \|\hat{\boldsymbol{\theta}}(\mathbf{v}) - \boldsymbol{\theta}^\star(\mathbf{v})\|^2$ with $\boldsymbol{\theta}^\star(\mathbf{v})$ the conditional mean parameter of a tile with descriptor $\mathbf{v}$. Source and target descriptors follow different distributions: the purpose of transfer is to serve cities unlike those measured, so covariate shift [30, 31] is the defining feature of the problem. In the *few-shot* variant, k target tiles are surveyed and their labels revealed before prediction. The operator is judged at two levels: the parameter error on the target tiles and, more importantly for planning, the error of the network quantities of Section 3.1 computed from the predicted parameters and compared with the ray-traced network.

3.3. Risk of label-averaging operators

Many transfer operators, including the Gaussian-kernel operator of [11] and k-nearest-neighbour averaging, predict a convex combination of source labels, $\hat{\boldsymbol{\theta}}_u = \sum_j w_j(\mathbf{v}_u)\boldsymbol{\theta}_j$ with $w_j \geq 0$, $\sum_j w_j = 1$ and weights that depend on descriptors only. Let the source labels be $\boldsymbol{\theta}_j = \boldsymbol{\theta}^\star(\mathbf{v}_j) + \boldsymbol{\eta}_j$, with $\boldsymbol{\eta}_j$ zero-mean label noise of covariance Σ_η, independent across j and of the descriptors. The target risk then splits into a smoothing bias and a noise-transmission term,

$$R = \underbrace{\mathbb{E}_{\mathbf{v}} \left\| \textstyle\sum_j w_j(\mathbf{v}) \boldsymbol{\theta}^\star(\mathbf{v}_j) - \boldsymbol{\theta}^\star(\mathbf{v}) \right\|^2}_{\text{smoothing bias}} + \underbrace{\operatorname{tr}(\Sigma_\eta)\, \mathbb{E}_{\mathbf{v}} \textstyle\sum_j w_j(\mathbf{v})^2}_{\text{noise transmission}} . \tag{7}$$

Proof. With $\hat{\boldsymbol{\theta}}(\mathbf{v}) - \boldsymbol{\theta}^\star(\mathbf{v}) = [\sum_j w_j \boldsymbol{\theta}^\star(\mathbf{v}_j) - \boldsymbol{\theta}^\star(\mathbf{v})] + \sum_j w_j \boldsymbol{\eta}_j$, the weights depend on descriptors only and $\mathbb{E}[\boldsymbol{\eta}_j] = \mathbf{0}$ independently of the descriptors, so the cross term has zero expectation; independence across j gives $\mathbb{E}\|\sum_j w_j \boldsymbol{\eta}_j\|^2 = \operatorname{tr}(\Sigma_\eta) \sum_j w_j^2$. □

The second term equals $\operatorname{tr}(\Sigma_\eta)\, \mathbb{E}[1/n_{\text{eff}}]$ with the effective neighbourhood size $n_{\text{eff}}(\mathbf{v}) = 1/\sum_j w_j(\mathbf{v})^2$. Concentrated weights lower the bias and raise the noise transmission; diffuse weights do the opposite. Under covariate shift the bias term grows, because the source set may contain no tile whose conditional mean resembles the target's; with noisy labels the second term matters as well. Ridge regression is also linear in the labels, and so is a regression tree once its partition is fixed; the difference is that their weights are not confined to the probability simplex and are shaped by the descriptor–label relation rather than fixed by an isotropic distance, which is the distinction that matters below.

4. Methodology

The evaluation rests on three layers, shown in Fig. 1: a data layer that builds geo-referenced scenes from open building data and ray traces them into corpora of links; a model layer that derives tile labels, descriptors, transfer operators and a site-aware link model from those corpora; and a decision layer that turns every predicted path-loss map into the serving sites, SINR, coverage, cell edges, site plans and coverage certificates a planner uses. Figure 2 gives the processing order and Fig. 3 the leave-one-city-out protocol under which every model is fitted and scored. Every stage is implemented in the accompanying repository and executed by a single command, and every number reported in Sections 5–6 is generated from its outputs.

4.1. Ray-traced corpora

The corpora are the ground truth of the study and are built so that every step from the building data to the links can be reproduced exactly. Each city is a square of side 3 km centred on a fixed coordinate and expressed in a local metric frame, the UTM projection of its zone shifted to the centre [38]. Building footprints are taken from the buildings theme of Overture Maps [39], release 2026-09-23.0, a versioned open dataset whose primary source is OpenStreetMap [40]; a pinned release makes the geometry exactly reproducible, which a live map query is not. Because grid north and true north differ, the footprint query box covers all four corners of the square. Building heights are resolved in order from the height attribute, from the number of floors at 3.2 m per floor, and otherwise from the median of the resolved heights of the city; the share of each source is reported per city. Footprint exteriors are clipped to the square, parts smaller than 4 m^2 are discarded, and the remainder is extruded to level-of-detail 1 and placed on a flat ground plane. Buildings use the ITU concrete model and the ground the ITU medium-dry-ground model [41].

Each city has 20 sites on rooftops: local height maxima of the building mesh on a 25 m grid are ranked by height and selected greedily with a minimum separation of 250 m, and the antenna is mounted 6 m above the roof; if too few candidates satisfy the separation, the remaining sites are drawn at random from the other rooftop candidates with a fixed seed. Ranking rooftops by height follows macro-cell practice but places the sites of high-rise cities on their tallest buildings: the median site height per city ranges from 38 m (Munich) to 318 m (New York), with a maximum of 478 m in New York (Table 5). Radio maps are computed with Sionna RT [7, 14] (version 2.1.0) at 3.5 GHz with isotropic antennas, 1×10^8 rays per site, maximum interaction depth 5, specular and diffuse reflection, refraction and diffraction (including edge diffraction) enabled, on a 10 m grid at 1.5 m receiver height. Cells with path loss above $L_{\max} = 150$ dB, beyond any practical coupling loss, are discarded; a cell that no site reaches below $L_{\max}$ is a coverage hole of the truth and is kept as such in the evaluation. The solver version and settings are stored with every city. Twenty sites on 9 km^2 give one site per 0.45 km^2, the density of a hexagonal layout with an inter-site distance of about 720 m, sparser than

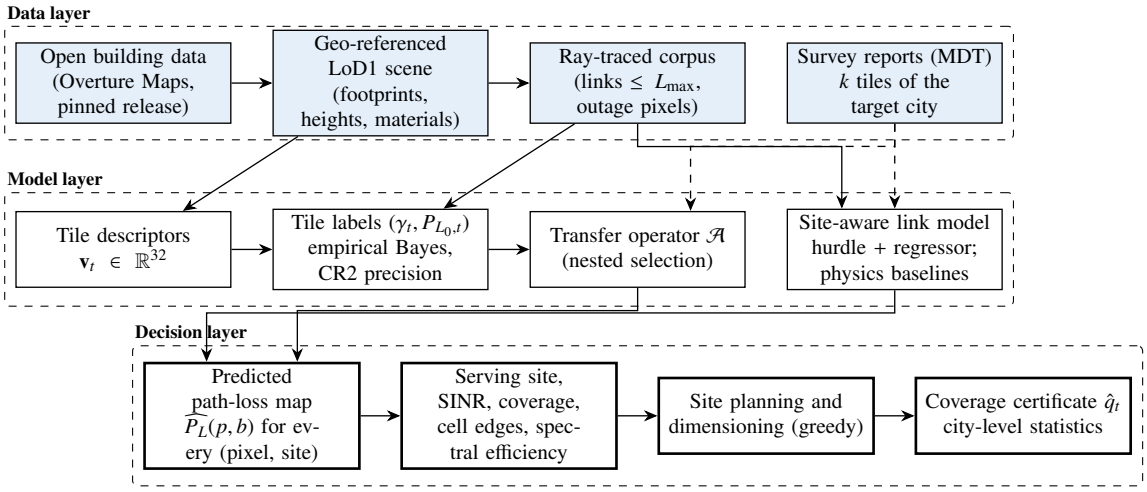


Figure 1: Architecture of the propagation-model layer of a digital twin network as evaluated in this study. Data layer: open building data, geo-referenced scenes and ray-traced corpora, plus optional survey reports from the target city. Model layer: tile descriptors and labels, transfer operators and the site-aware link model. Decision layer: the predicted path-loss map and the network quantities, site plans and certificates derived from it. Dashed arrows mark the optional survey information.

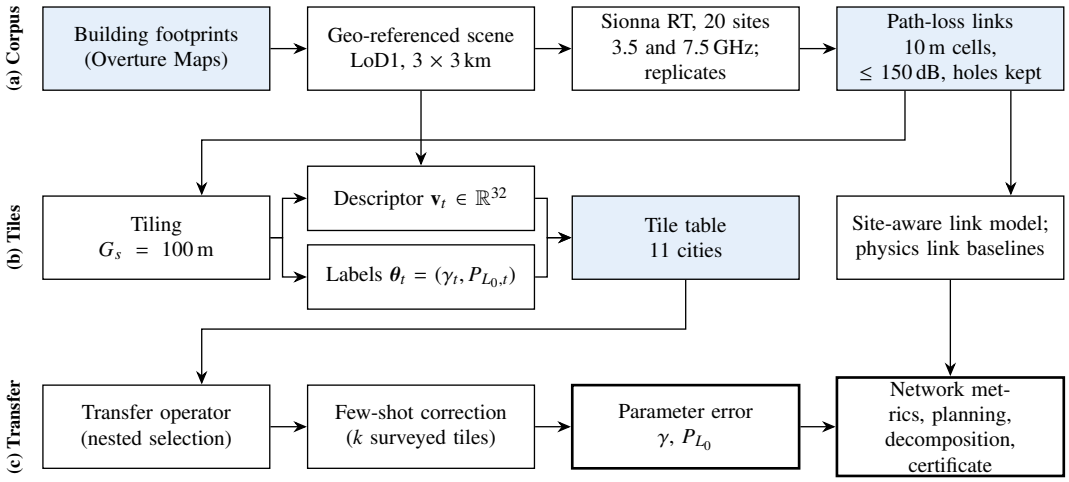


Figure 2: Processing pipeline. (a) Geo-referenced scenes are built from open building footprints and ray traced. (b) Links are grouped into tiles; each tile receives a label $\boldsymbol{\theta}_t = (\gamma_t, P_{L_0,t})$ and a descriptor $\mathbf{v}_t$ computed from the same scene geometry. (c) Tile transfer operators fitted on source cities predict the labels of a held-out city, optionally corrected with k surveyed tiles; the site-aware link model and the per-link physics baselines predict every link from its geometry. All are evaluated through the network quantities of Section 3.1, a site plan and a coverage certificate.

the 500 m of the 3GPP urban-macro scenario [9]. The density and the siting rule are tested in Section 5.8.

Four further corpora use the same scenes. The *second band* re-traces every site at 7.5 GHz, in the 7 GHz to 24 GHz range studied for 6G [5]. The *random-siting replicate* re-traces 4 cities (Amsterdam, London, New York, Paris) at 3.5 GHz with the same number of sites taken from the rooftop candidates in random order instead of by height, under the same separation rule; the *mid-rise replicate* draws them in random order from the rooftops between 12 m and 45 m above ground, the site-height range of the 3GPP urban-macro scenario; and the *materials replicate* keeps the main siting but assigns each building part one of 5 ITU materials (brick, concrete, marble, metal, wood) at random with a fixed seed instead of concrete for all. The geometry, and hence every descriptor and link feature, is identical across these replicates; only the ray-traced truth changes.

The planar radio-map estimator of the solver weights a ray crossing the measurement plane by the inverse cosine of its incidence angle, and a diffracted ray by an integration weight over the diffraction cone; the first weight is unbounded and the second can be very large, so a cell occasionally receives one overweighted contribution. Its path loss can then fall below free space by tens of decibels, which no incoherent sum of a few propagation paths allows. With the free-space loss

$$P_{L,\mathrm{FS}}(d) = 20 \log_{10} \frac{4\pi d f}{c_0} \tag{8}$$

at carrier frequency f and speed of light c_0, every link with $P_L(p,b) < P_{L,\mathrm{FS}}(d_{pb}) - 6\,\mathrm{dB}$ is removed: 1,170 links, 0.013% of the corpus. For the site with the most such links, disabling diffraction reduces the links more than 6 dB below free space from 69 to 0 with otherwise identical settings, which identifies

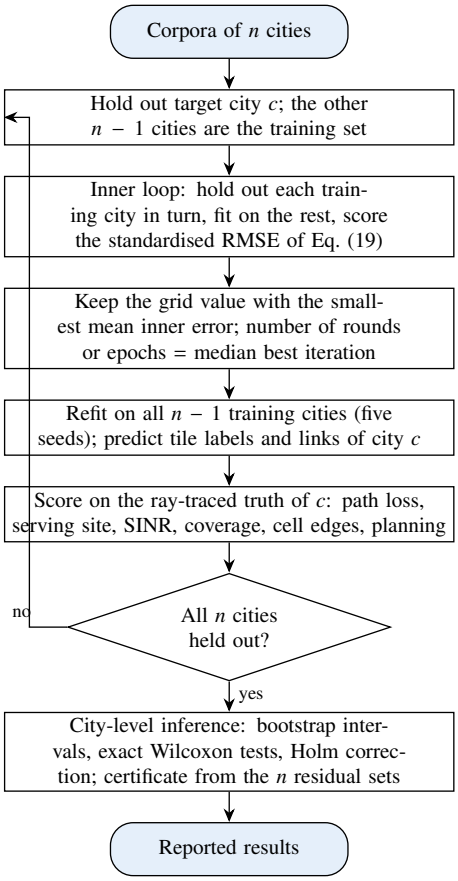


Figure 3: Nested leave-one-city-out protocol. Every hyper-parameter is chosen inside the training cities; the held-out city is used only for scoring.

the diffraction weights as their source.

4.2. Tile labels and descriptors

The labels are the quantity that tile transfer carries from city to city, and their precision bounds what any operator can achieve; this subsection therefore derives the estimator and three estimates of its error. Links are assigned to tiles of side $G_s = 100\,\mathrm{m}$ by their receiver position; tiles with fewer than 30 links are not labelled. With n_t links in tile t, centred regressor $x_i - \bar{x}_t$ and the sums $S_{xx,t} = \sum_i (x_i - \bar{x}_t)^2$ and $S_{xy,t} = \sum_i (x_i - \bar{x}_t)(y_i - \bar{y}_t)$, the ordinary least-squares (OLS) slope and its variance under independent residuals of variance s_t^2 are

$$b_t = \frac{S_{xy,t}}{S_{xx,t}}, \qquad v_t^{\mathrm{iid}} = \frac{s_t^2}{S_{xx,t}} = \frac{s_t^2}{n_t \,\mathrm{Var}_t(x)}. \tag{9}$$

The slope is poorly determined because every receiver sees the serving sites from a similar distance: a receiver at distance d from its site sees a tile of side G_s span only $\Delta x \approx 10\log_{10}(1 + G_s/d)$, which is 0.41 dB for $G_s = 100\,\mathrm{m}$ and $d = 1\,\mathrm{km}$, so $\mathrm{Var}_t(x)$ vanishes and v_t^{iid} diverges. The slope is therefore shrunk towards the city-wide mean with weights set by the data (empirical Bayes [42]).

Proposition 1 (Empirical-Bayes shrinkage)**.** *Let b_t be the OLS slope of tile t, computed with x centred inside the tile, with sampling variance v_t, and let the true slopes vary between tiles with mean μ and variance τ^2. Among estimators $\hat{\gamma}_t = (1-w)b_t + w\mu$ the mean squared error $(1-w)^2 v_t + w^2\tau^2$ is minimised by*

$$w_t = \frac{v_t}{v_t + \tau^2} = \frac{\lambda_t}{\lambda_t + S_{xx,t}}, \qquad \lambda_t = \frac{S_{xx,t}\, v_t}{\tau^2}, \tag{10}$$

which is ridge regression on the centred slope with penalty λ_t and prior mean μ. Its risk is $v_t\tau^2/(v_t + \tau^2) \le \min(v_t, \tau^2)$, bounded by τ^2 however degenerate the tile geometry.

Proof. Given the true slope, b_t has mean γ_t and variance v_t, and the true slopes have mean μ and variance τ^2. For fixed w, $\hat{\gamma}_t - \gamma_t = (1-w)(b_t - \gamma_t) + w(\mu - \gamma_t)$; the two terms are uncorrelated, so the mean squared error is $(1-w)^2 v_t + w^2\tau^2$, a convex quadratic in w minimised at $w_t = v_t/(v_t + \tau^2)$ with value $v_t\tau^2/(v_t + \tau^2)$, which is at most $\min(v_t, \tau^2)$. With centred x, the ridge problem $\min_\gamma \sum_i (y_i - \bar{y} - \gamma x_i)^2 + \lambda(\gamma - \mu)^2$ has solution $(S_{xy} + \lambda\mu)/(S_{xx} + \lambda) = (1-w)b_t + w\mu$ with $w = \lambda/(\lambda + S_{xx})$; setting $\lambda_t = S_{xx} v_t/\tau^2$ gives $w = w_t$. □

The prior parameters (μ, τ^2) are estimated per city from the tiles with a non-degenerate span by the DerSimonian–Laird moment estimator [43]: with $\omega_t = 1/v_t$, the fixed-effect mean $\mu_{\mathrm{FE}} = \sum_t \omega_t b_t / \sum_t \omega_t$ and the heterogeneity statistic $Q = \sum_t \omega_t (b_t - \mu_{\mathrm{FE}})^2$ over m tiles,

$$\hat{\tau}^2 = \max\Big\{0,\ \frac{Q - (m-1)}{\sum_t \omega_t - \sum_t \omega_t^2 / \sum_t \omega_t}\Big\}, \qquad \hat{\mu} = \frac{\sum_t b_t/(v_t + \hat{\tau}^2)}{\sum_t 1/(v_t + \hat{\tau}^2)}. \tag{11}$$

The label is then $\hat{\gamma}_t = (1 - w_t)b_t + w_t\hat{\mu}$ with the intercept left unpenalised, $\hat{P}_{L_0,t} = \bar{y}_t - \hat{\gamma}_t \bar{x}_t$; a tile with no span takes $\hat{\gamma}_t = \hat{\mu}$. The shrinkage weight w_t is published with every label. Algorithm 1 states the procedure; it is vectorised over all links of a city.

Algorithm 1: Tile labels

Input: links (x_i, y_i, t_i, b_i) of one city; minimum count $n_{\min}$
Output: $\hat{\theta}_t$, w_t, iid and cluster-robust standard errors

foreach *tile t with* $n_t \ge n_{\min}$ **do**
  $\bar{x}_t, \bar{y}_t, S_{xx,t}, S_{xy,t}$; $b_t \leftarrow S_{xy,t}/S_{xx,t}$; $v_t \leftarrow$ cluster-robust variance of b_t (Eq. (12))
end
$(\hat{\mu}, \hat{\tau}^2) \leftarrow$ DerSimonian–Laird over $\{(b_t, v_t)\}$ (Eq. (11))
foreach *tile t* **do**
  $w_t \leftarrow v_t/(v_t + \hat{\tau}^2)$; $\hat{\gamma}_t \leftarrow (1 - w_t)b_t + w_t\hat{\mu}$; $\hat{P}_{L_0,t} \leftarrow \bar{y}_t - \hat{\gamma}_t \bar{x}_t$
  residuals $e_i \leftarrow y_i - \hat{P}_{L_0,t} - \hat{\gamma}_t x_i$
  standard errors from Eq. (12) with iid and with site clusters
end

Radio-map cells served by the same site share propagation paths, so the independent-residual variance of Eq. (9) understates v_t. The estimator is linear in the path losses: in the centred parametrisation $y_i = a_t + \gamma_t(x_i - \bar{x}_t) + e_i$ with design rows $\mathbf{z}_i = (1, x_i - \bar{x}_t)$, the coefficients are $\mathbf{M}\sum_i \mathbf{z}_i y_i$ up to the prior term, with $\mathbf{M} = \mathrm{diag}(1/n_t,\ 1/(S_{xx,t} + \lambda_t))$ ($\lambda_t = 0$ for the OLS slope). With the links of the tile partitioned by serving site into

G clusters $\mathcal{I}_1, \dots, \mathcal{I}_G$, with design blocks $\mathbf{Z}_c$ and residuals $\mathbf{e}_c$, the cluster-robust variance is

$$\widehat{\mathrm{Var}}_{\mathrm{cl}} = \mathbf{M}\Big(\sum_{c=1}^{G} \mathbf{u}_c \mathbf{u}_c^{\top}\Big)\mathbf{M}, \quad \mathbf{u}_c = \mathbf{Z}_c^{\top}(\mathbf{I} - \mathbf{H}_{cc})^{-1/2}\mathbf{e}_c, \qquad (12)$$

with $\mathbf{H}_{cc} = \mathbf{Z}_c \mathbf{M} \mathbf{Z}_c^{\top}$ (Moore–Penrose inverse root where $\mathbf{I} - \mathbf{H}_{cc}$ is singular). This is the bias-reduced linearisation (CR2) of [44, 45]; without the factor $(\mathbf{I} - \mathbf{H}_{cc})^{-1/2}$ it is the usual estimator [46, 47], which is biased downwards when the clusters are few, as here. Since $\mathbf{Z}_c^{\top} f(\mathbf{Z}_c \mathbf{M} \mathbf{Z}_c^{\top}) = f(\mathbf{Z}_c^{\top}\mathbf{Z}_c \mathbf{M})\mathbf{Z}_c^{\top}$ for $f(s) = (1 - s)^{-1/2}$, every $\mathbf{u}_c$ requires only the eigen-decomposition of a 2×2 matrix. The cluster-robust variance of the slope is used as v_t in Eq. (10), never below the independent value; a tile served by a single site uses the independent value multiplied by the median ratio of the two over the city (its design effect). Because the number of clusters per tile is small, Eq. (12) is complemented by a split-half estimate: the sites of a city are divided at random into two disjoint halves, the complete estimator (including its own $\hat{\mu}, \hat{\tau}^2$) is refitted on each half, and

$$\widehat{\mathrm{se}}_{\mathrm{split}}(\hat{\gamma}_t) = \frac{\mathrm{sd}(\hat{\gamma}_t^{(1)} - \hat{\gamma}_t^{(2)})}{2} \qquad (13)$$

estimates the standard error of the full-data label: the standard deviation of the difference divided by $\sqrt{2}$ is the error of a half-data label, and a further division by $\sqrt{2}$ extrapolates to the full data under the assumption that the variance scales with the inverse number of sites. Five random partitions are pooled.

The descriptor must describe exactly the geometry the ray tracer saw. It is therefore computed from the same footprint parts, heights and scene clipping as the extruded mesh, with every part clipped to the tile it falls in. It contains the 32 built-environment features of [11]: footprint ratio and building count in five height classes (< 5, 5–10, 10–30, 30–60, ≥ 60 m); mean and standard deviation of footprint area, height and volume; total footprint density; and, for horizontal planes at 5, 10, 30, 60 and 100 m, the number of buildings reaching the plane and the mean and standard deviation of the distance between their centroids. The topographic, land-cover and climatic descriptors of [11] are omitted on purpose: the scenes have a flat ground, no water or vegetation surfaces and no weather, so these features would describe nothing the solver modelled and could only act as city identifiers. Descriptors are stored unnormalised; every fit scales them by the column-wise maximum absolute value of its own training tiles,

$$\tilde{v}_{t,\ell} = \frac{v_{t,\ell}}{\max_{j \in \mathcal{T}_{\mathrm{S}}} |v_{j,\ell}|}, \qquad \ell = 1, \dots, p, \qquad (14)$$

never of the target.

4.3. Transfer operators and nested selection

Thirteen operators are compared under one protocol: constant predictors, label-averaging operators and learned regressors, with every hyper-parameter chosen inside the training cities. Their prediction rules are the following.

- *Source median*: the median source label, the no-information reference.
- *3GPP UMa*: the line-of-sight and non-line-of-sight single-slope formulas of TR 38.901 [9] at the reference distance, which give fixed parameters for every tile, and *COST-231 Hata* [48] (metropolitan correction, median site height of the source cities), used outside its validity range of 1.5 GHz to 2 GHz and 1 km to 20 km as a further standard model.
- *Kernel (Nadaraya–Watson)* [49, 50], the operator of [11]:

$$\hat{\boldsymbol{\theta}}_u = \sum_j \tilde{w}_j(u)\boldsymbol{\theta}_j, \qquad \tilde{w}_j(u) \propto \exp\Big(-\frac{\|\mathbf{v}_u - \mathbf{v}_j\|^2}{2\sigma^2}\Big), \qquad (15)$$

  computed with the minimum squared distance subtracted in the exponent, which leaves the weights unchanged and avoids numerical underflow.
- *GP-ARD*: a Gaussian process with one length scale per descriptor (automatic relevance determination) [51], i.e. a kernel operator whose metric is learned. With the kernel

$$k(\mathbf{v}, \mathbf{v}') = \sigma_f^2 \exp\Big(-\tfrac{1}{2}\sum_{\ell=1}^{p} \frac{(v_\ell - v'_\ell)^2}{\ell_\ell^2}\Big) + \sigma_n^2 \mathbb{1}\{\mathbf{v} = \mathbf{v}'\}, \qquad (16)$$

  the posterior mean of each standardised target is $\hat{\theta}_u = \mathbf{k}_u^{\top}\mathbf{K}^{-1}\mathbf{z}$, where $\mathbf{K}$ is the kernel matrix of the source tiles, $\mathbf{k}_u$ the vector of kernel values between $\mathbf{v}_u$ and the source tiles and $\mathbf{z}$ the standardised source labels; the hyper-parameters $(\sigma_f, \ell_1, \dots, \ell_p, \sigma_n)$ maximise the marginal likelihood on 600 random source tiles and the posterior mean uses every source tile.
- *k-NN*: the mean label of the k nearest source tiles, $\hat{\boldsymbol{\theta}}_u = k^{-1}\sum_{j \in \mathcal{N}_k(u)} \boldsymbol{\theta}_j$.
- *Ridge*: with $\mathbf{V}$ the $N \times p$ matrix of scaled source descriptors and $\boldsymbol{\Theta}$ the $N \times 2$ matrix of labels,

$$\hat{\boldsymbol{\theta}}_u^{\top} = \tilde{\mathbf{v}}_u^{\top}(\mathbf{V}^{\top}\mathbf{V} + \alpha\mathbf{I})^{-1}\mathbf{V}^{\top}\boldsymbol{\Theta}, \qquad (17)$$

  with an unpenalised intercept; a two-layer *MLP* (two hidden layers of 256 units, targets standardised on the training tiles, trained epoch by epoch); gradient-boosted trees (*GBDT*, LightGBM [52]) with one model per target, each an additive ensemble $F_M(\mathbf{v}) = \sum_{m=1}^{M} f_m(\mathbf{v})$ of regression trees fitted by gradient boosting of the squared error; and a *neural encoder* with a constrained (γ, P_{L_0}) head trained with a heteroscedastic Gaussian likelihood.
- Two domain-adaptation variants that use the unlabelled descriptors of the target city. *GBDT-IW* weights the source tiles by the density ratio of target to source descriptors estimated with a logistic domain classifier [30]: with π_j the classifier's probability that source tile j belongs to the target and N_{T} target tiles,

$$w_j = \min\Big\{\frac{\pi_j}{1 - \pi_j}\frac{N}{N_{\mathrm{T}}},\ q_{0.99}\Big\}, \qquad (18)$$

clipped at the 99th percentile $q_{0.99}$ of the weights and normalised to unit mean. *CORAL* re-colours the source descriptors with the target covariance before ridge regression [32], $\mathbf{V}_S\mathbf{C}_S^{-1/2}\mathbf{C}_T^{1/2}$, with both covariances regularised by a small multiple of the identity.

Every hyper-parameter (bandwidth σ, k, ridge, CORAL and MLP penalties, number of boosting rounds, number of epochs) is chosen inside the training cities by an inner leave-one-training-city-out loop (Fig. 3): each training city is held out once, the operator is fitted on the others and scored by the root-mean-square standardised error over both parameters,

$$\ell = \Big(\frac{1}{2n_v}\sum_{i=1}^{n_v}\sum_{q\in\{\gamma,P_{L_0}\}}\frac{(\hat\theta_{i,q}-\theta_{i,q})^2}{s_q^2}\Big)^{1/2}, \tag{19}$$

with s_q the standard deviation of target q over the training tiles, so that the exponent and the intercept count equally, and the grid value with the smallest mean inner error is kept. The operator is then refitted on all training cities; for the operators with early stopping, the number of rounds or epochs of the final fit is the median best iteration over the inner folds. The two expensive operators, the neural encoder and the link model of Section 4.5, use 3 inner validation cities drawn with a fixed seed. No city is ever reserved for validation, so the procedure is symmetric in the source cities, which the certificate of Section 4.6 requires, and the held-out city is never touched. The chosen values are listed in Table 8. Stochastic operators are run with five seeds; the inner selection uses seed 0.

The behaviour of the kernel operator under shift can be characterised analytically. Write $d_j = \|\mathbf{v}_u - \mathbf{v}_j\|^2$, $\delta_j = d_j - \min_k d_k \in [0, D_u]$, $\bar{\boldsymbol\theta} = N^{-1}\sum_j \boldsymbol\theta_j$ and $\sigma_\theta^2 = N^{-1}\sum_j\|\boldsymbol\theta_j - \bar{\boldsymbol\theta}\|^2$.

Proposition 2 (Two regimes of kernel transfer)**.** *(a) If $\epsilon_u = D_u/(2\sigma^2) < 1$, then*

$$\hat{\boldsymbol\theta}_u = \bar{\boldsymbol\theta} - \frac{1}{2\sigma^2}\operatorname{Cov}_j(\delta_j, \boldsymbol\theta_j) + O(\epsilon_u^2), \quad \big\|\hat{\boldsymbol\theta}_u - \bar{\boldsymbol\theta}\big\| \le \frac{D_u\,\sigma_\theta}{4\sigma^2} + O(\epsilon_u^2), \tag{20}$$

and $n_{\mathrm{eff}}(u) = N(1 + O(\epsilon_u^2))$: the operator returns the source mean up to a first-order correction. (b) Let g_u be the gap between the smallest and second-smallest squared distance. Then the weight of the nearest source tile satisfies

$$\tilde w_{(1)} \ge \big(1 + (N-1)e^{-g_u/(2\sigma^2)}\big)^{-1}, \tag{21}$$

so $\hat{\boldsymbol\theta}_u$ tends to the label of the nearest source tile when $g_u \gg 2\sigma^2 \ln N$.

Proof. (a) Put $\alpha = 1/(2\sigma^2)$. The factor $e^{-\alpha\min_k d_k}$ cancels in Eq. (15), so $\tilde w_j = e^{-\alpha\delta_j}/\sum_k e^{-\alpha\delta_k}$ with $\alpha\delta_j \le \epsilon_u < 1$. By Taylor's theorem $e^{-\alpha\delta_j} = 1 - \alpha\delta_j + r_j$ with $0 \le r_j \le \epsilon_u^2/2$, hence $\tilde w_j = N^{-1}(1 - \alpha(\delta_j - \bar\delta)) + O(\epsilon_u^2/N)$ and $\hat{\boldsymbol\theta}_u = \bar{\boldsymbol\theta} - \alpha\operatorname{Cov}_j(\delta_j, \boldsymbol\theta_j) + O(\epsilon_u^2)$. By the Cauchy–Schwarz inequality $\|\operatorname{Cov}_j(\delta_j, \boldsymbol\theta_j)\| \le \sigma_\delta\sigma_\theta$ with $\sigma_\delta \le D_u/2$, which gives the bound; the expansion of the weights gives $\sum_j \tilde w_j^2 = N^{-1}(1 + O(\epsilon_u^2))$. (b) With the nearest tile at index (1), every other weight satisfies $e^{-\alpha\delta_j} \le e^{-\alpha g_u}$, so $\tilde w_{(1)} = 1/(1 + \sum_{j\ne(1)} e^{-\alpha\delta_j}) \ge 1/(1 + (N-1)e^{-\alpha g_u})$. □

The minimum distance cancels from Eq. (15); a target far from every source tile is therefore *not* by itself pushed to either regime. Which regime applies is decided by the selected bandwidth relative to the spread D_u and the gap g_u; it is measured (Section 5.3) rather than assumed. Both regimes carry a failure mode under shift: the flat regime returns the source mean and ignores the target descriptor; the nearest-neighbour regime copies one source tile and transmits its label noise undamped (Eq. (7) with $n_{\mathrm{eff}} = 1$).

4.4. Few-shot correction and survey design

When a small survey of the target city is available, the zero-shot prediction can be corrected; the correction must respect the precision of the surveyed labels and must not be applied where the inner cities show no systematic error. With k target tiles surveyed, the zero-shot prediction $g(\mathbf{v})$ is corrected by the residuals $r_i = \boldsymbol\theta_i - g(\mathbf{v}_i)$ of the k anchors, weighting each anchor by its precision and shrinking the result by empirical Bayes, per parameter:

$$w_i = \frac{1}{\mathrm{se}_i^2 + s^2}, \quad c = \frac{\sum_i w_i r_i}{\sum_i w_i}, \quad V = \frac{1}{\sum_i w_i}, \quad c_{\mathrm{EB}} = \frac{\tau_b^2}{\tau_b^2 + V}\,c, \tag{22}$$

with se_i the cluster-robust standard error of the anchor label. The within-city error variance s^2 of the operator and the between-city variance τ_b^2 of its zero-shot bias are estimated on the inner validation cities of the fold, without any target information: with $\bar e_c$ and RMSE_c the mean error and root-mean-square error of the operator on inner city c,

$$\hat s^2 = \operatorname{mean}_c(\mathrm{RMSE}_c^2 - \bar e_c^2), \qquad \hat\tau_b^2 = \operatorname{Var}_c(\bar e_c). \tag{23}$$

When the inner cities show no systematic bias, $\hat\tau_b^2 \to 0$ and the correction vanishes; when they do, the correction is the precision-weighted mean residual. It is compared with the plain mean residual (*bias*) and with a precision-weighted ridge model of the residual on the descriptors (*residual*),

$$\hat{\boldsymbol\beta} = \arg\min_{\boldsymbol\beta}\sum_{i=1}^{k} w_i(r_i - \boldsymbol\beta^\top\tilde{\mathbf{v}}_i)^2 + \lambda\|\boldsymbol\beta\|^2, \tag{24}$$

with $\lambda = 10$ on normalised descriptors. Anchors are drawn uniformly at random, as the k tiles closest to a random tile (*contiguous*), which mimics a single drive-test area [17], or by a *design* that uses no measurement: the k tiles that greedily maximise the facility-location objective

$$F_{\mathrm{fl}}(T) = \sum_{i\in\mathrm{target}}\max_{j\in T}\exp\Big(-\frac{\|\mathbf{v}_i - \mathbf{v}_j\|^2}{2h^2}\Big) \tag{25}$$

over the target tiles, with h the median pairwise descriptor distance; the objective is monotone submodular, so the greedy set is within a factor $(1 - 1/e)$ of the optimum [16, 19]. Evaluation is always on the unsurveyed tiles, anchors are shared by all operators, and ten random draws are made per $k \in \{3, 5, 10, 25, 50, 100\}$.

For the link model of Section 4.5, the links observed in k surveyed tiles, which carry their serving site as MDT reports

do [17, 18], calibrate a per-site offset. Model the residual of a link of site s as $o_s + \varepsilon$, with independent site offsets o_s of prior mean $\bar{o}$ and variance τ_o^2 and link noise of variance σ_ε^2, and let $\kappa = \sigma_\varepsilon^2/\tau_o^2$. With n_s observed links of site s and mean residual $\bar{r}_s$, the posterior mean and variance of o_s are

$$\hat{o}_s = \frac{n_s \bar{r}_s + \kappa \bar{o}}{n_s + \kappa}, \qquad \mathrm{Var}(o_s \mid \text{survey}) = \frac{\tau_o^2 \kappa}{n_s + \kappa}, \tag{26}$$

which are used with $\bar{o}$ the mean residual of all observed links. The prior strength κ is estimated from the surveyed links themselves (empirical Bayes): $\hat{\sigma}_\varepsilon^2$ is the pooled within-site variance of the residuals and $\hat{\tau}_o^2$ the DerSimonian–Laird between-site variance of Eq. (11) applied to the site means with variances $\hat{\sigma}_\varepsilon^2/n_s$, so that a survey whose site means scatter no more than their sampling error applies the overall offset only; a fixed $\kappa = 50$ links is reported as an ablation.

Proposition 3 (Survey design)**.** *Let $c_{ts} \geq 0$ be the number of links of site s observable in tile t and $n_s(T) = \sum_{t \in T} c_{ts}$ for a set T of surveyed tiles. The total reduction of posterior variance of the site offsets,*

$$F(T) = \tau_o^2 \sum_s \frac{n_s(T)}{n_s(T) + \kappa}, \tag{27}$$

is monotone and submodular in T, so the greedy choice of k tiles attains $F(T_{\text{greedy}}) \geq (1 - 1/e) \max_{|T|=k} F(T)$.

Proof. From Eq. (26), $\tau_o^2 - \mathrm{Var}(o_s \mid T) = \tau_o^2 n_s(T)/(n_s(T) + \kappa)$, so F is the total variance reduction. Each $n_s(T)$ is modular and non-decreasing in T, and $x \mapsto x/(x + \kappa)$ is non-decreasing and concave on $x \geq 0$; a non-decreasing concave function of a non-negative modular function is monotone submodular, and a non-negative sum of such functions is as well. The greedy guarantee is that of [16]. □

Before any measurement c_{ts} is unknown; the *variance design* uses the number of links of site s in tile t that the hurdle classifier of the zero-shot link model predicts below the cut-off. Because maximising F alone favours tiles that see many sites, which need not be representative of the city, a *stratified design* first partitions the tiles into k descriptor strata (k-means) and takes in each stratum the tile with the largest marginal gain of F. Both are compared with random surveys for $k \in \{10, 25, 50\}$ (ten draws). The surveyed pixels keep their model prediction, and every few-shot metric is computed on the unsurveyed pixels only.

4.5. Site-aware link model and per-link baselines

As a transfer model free of the receiver-tile restrictions, the path loss of each (pixel, site) link is predicted directly from the geometry along the link, which is available for a city without any measurement. The building heights are rasterised at 2 m and the straight segment from the site to the pixel is sampled at 96 interior points, where the building height $H(s)$ is compared with the line-of-sight height $h_{\text{los}}(s)$. The link features are the 3-D and horizontal distances, the site height and the elevation angle; the fraction of the segment with $H > h_{\text{los}}$, the number of separate obstructions and the length over footprints; the largest excess height $\max_s(H - h_{\text{los}})$; the single knife-edge loss of the most obstructing point, which for an edge of height h above the line of sight at distances d_1 and d_2 from the two ends is

$$J(\nu) = 6.9 + 20 \log_{10}\Big(\sqrt{(\nu - 0.1)^2 + 1} + \nu - 0.1 \Big), \qquad \nu = h \sqrt{\frac{2}{\lambda}\Big(\frac{1}{d_1} + \frac{1}{d_2}\Big)}, \tag{28}$$

for $\nu > -0.78$ and zero otherwise [13]; whether the receiver lies inside a footprint and that building's height; and the built fraction and mean building height in a 100 m square window centred on the receiver and the built fraction in a 200 m window centred on the site. The 32 descriptors of the receiver's tile are appended, giving 46 features in total.

The ray tracer reports only links below the cut-off, so a regressor trained on them never learns that a distant, obstructed site is not received at all; it would predict a finite loss for every link and overstate the reach of every site. A hurdle model is therefore used: a gradient-boosted classifier [52] of whether a (pixel, site) pair lies below the cut-off, trained on 120,000 pairs of each training city (all sites and the pixels of its labelled tiles, including the pairs beyond the cut-off), and a gradient-boosted regressor of the path loss, trained on 120,000 ray-traced links of each training city. With $\hat{\pi}(p, b)$ the classifier's probability and $\widehat{P_{L\text{reg}}}(p, b)$ the regression, the predicted map is

$$\widehat{P_L}(p, b) = \begin{cases} \widehat{P_{L\text{reg}}}(p, b), & \hat{\pi}(p, b) \geq \tfrac{1}{2}, \\ +\infty, & \text{otherwise}, \end{cases} \tag{29}$$

where $+\infty$ denotes zero received power. The number of rounds of both models follows the nested rule of Section 4.3 (learning rate 0.1, 127 leaves, at most 1500 rounds, five seeds). The path-loss error of the link model is that of the regressor over the ray-traced links; the regressor alone, which keeps every link, is reported as an ablation, and a censored error that scores every pair with prediction and truth clipped at $L_{\max}$ is reported as well.

Three non-learned predictors use the same link geometry and the deterministic hurdle "predicted loss $\leq L_{\max}$"; they answer the question whether a learned link model is needed at all once the obstruction geometry is available. The first is free space plus the single knife-edge loss of Eq. (28). The second is free space plus the Deygout three-edge construction of ITU-R P.526 [13]: with ν_{p} the largest Fresnel parameter along the path (the principal edge) and ν_1, ν_2 the largest parameters on the two sub-paths from each end to the principal edge,

$$P_{L,\text{D}} = P_{L,\text{FS}}(d) + J(\nu_{\text{p}}) + \mathbb{1}\{\nu_{\text{p}} > -0.78\}[J(\nu_1) + J(\nu_2)]. \tag{30}$$

The third is the 3GPP UMa model of TR 38.901 [9] with the line-of-sight state of each link decided from the geometry: with d_3 and d_2 the 3-D and horizontal distances in metres, f_c in GHz, h_{BS} and h_{UT} the antenna heights and the breakpoint distance

$$d'_{\mathrm{BP}} = 4(h_{\mathrm{BS}}-1)(h_{\mathrm{UT}}-1)f/c_0,$$

$$\begin{aligned} P_{L,\mathrm{LOS}} &= 28.0 + 22\log_{10} d_3 + 20\log_{10} f_c && d_2 \le d'_{\mathrm{BP}},\\ P_{L,\mathrm{LOS}} &= 28.0 + 40\log_{10} d_3 + 20\log_{10} f_c \\ &\quad - 9\log_{10}(d'^2_{\mathrm{BP}} + (h_{\mathrm{BS}}-h_{\mathrm{UT}})^2) && d_2 > d'_{\mathrm{BP}},\\ P_{L,\mathrm{NLOS}} &= \max\{P_{L,\mathrm{LOS}},\ 13.54 + 39.08\log_{10} d_3 \\ &\quad + 20\log_{10} f_c - 0.6(h_{\mathrm{UT}}-1.5)\}. \end{aligned} \tag{31}$$

To attribute the gap between the tile model and link transfer, every tile operator is also evaluated (i) with the hurdle of Eq. (29) applied to its map (seed-matched), (ii) as a dual-slope model with one (γ, P_{L_0}) pair per tile fitted to the line-of-sight links and one to the obstructed links, the state of each (pixel, site) pair decided from the geometry, each pair falling back to the all-link label where fewer than 30 links of its kind exist, and (iii) with both. GBDT transfers the four labels of the dual-slope model under the nested rule.

4.6. Network-level evaluation, planning and certification

The decision layer scores every predicted map against the ray-traced truth through the quantities of Section 3.1. For the held-out city, each operator's tile parameters and the true site positions define a predicted path-loss map for every (pixel, site) pair through Eq. (4); the ray-traced map is the truth over every cell of the solver grid inside a labelled tile, with pairs beyond the cut-off treated as zero received power and cells no site reaches as outage pixels (4.8% of the pixels, 0.1–16.7% across cities). The link budget uses $P_{\mathrm{tx}} = 46\,\mathrm{dBm}$, 20 MHz bandwidth and a 7 dB noise figure ($P_{\mathrm{N}} = -94.0\,\mathrm{dBm}$). Let $\mathcal{L} = \{(p,b) : P_L(p,b) \le L_{\max}\}$ be the set of ray-traced links, $\widehat{\mathcal{L}}$ the set of pairs the model predicts below the cut-off, $\mathcal{P}_{\mathrm{r}}$ the pixels the truth reaches and $\hat{s}(p)$ the predicted serving site ($\hat{s}(p) = \varnothing$ where the model predicts no reaching site). The path-loss metrics are

$$\begin{aligned} \mathrm{RMSE} &= \Big(\frac{1}{|\mathcal{L}|}\sum_{(p,b)\in\mathcal{L}} \big(\widehat{P_L}(p,b) - P_L(p,b)\big)^2\Big)^{1/2},\\ \mathrm{RMSE_c} &= \Big(\frac{1}{|\mathcal{P}||\mathcal{B}|}\sum_{p,b}\big(\min\{\widehat{P_L}, L_{\max}\} - \min\{P_L, L_{\max}\}\big)^2\Big)^{1/2}, \end{aligned} \tag{32}$$

the first over the ray-traced links (with the regression part for a hurdle model) and the second, censored, over every pair; the link-existence and association metrics are

$$\begin{aligned} \mathrm{prec} &= \frac{|\widehat{\mathcal{L}}\cap\mathcal{L}|}{|\widehat{\mathcal{L}}|}, \qquad \mathrm{rec} = \frac{|\widehat{\mathcal{L}}\cap\mathcal{L}|}{|\mathcal{L}|},\\ A_{\mathrm{serv}} &= \frac{1}{|\mathcal{P}_{\mathrm{r}}|}\sum_{p\in\mathcal{P}_{\mathrm{r}}} \mathbb{1}\{\hat{s}(p) = s(p)\}, \end{aligned} \tag{33}$$

so that a pixel the model does not reach counts as wrongly associated, and the reach accuracy is the share of pixels whose reached state $\mathbb{1}\{\exists b : (p,b)\in\mathcal{L}\}$ is predicted correctly. Coverage is compared at $L \in \{120, 130, 140\}$ dB through the agreement $|\mathcal{P}|^{-1}\sum_p \mathbb{1}\{\hat{c}_L(p) = c_L(p)\}$ and the error $\hat{C}_L - C_L$ of the covered share; the SINR of Eq. (1), with values below -20 dB (including pixels at which no site is predicted to be received) set to -20 dB for the truth and every predictor alike, is compared through the mean absolute error and the Kolmogorov–Smirnov distance of the two empirical distributions,

$$\begin{aligned} \mathrm{MAE_{SINR}} &= \frac{1}{|\mathcal{P}|}\sum_p \big|\widehat{\mathrm{SINR}}(p) - \mathrm{SINR}(p)\big|,\\ D_{\mathrm{KS}} &= \sup_x \big|\hat{F}_{\widehat{\mathrm{SINR}}}(x) - \hat{F}_{\mathrm{SINR}}(x)\big|, \end{aligned} \tag{34}$$

and the spectral efficiency of Eq. (3) through the error of its mean. The cell edges are compared by the symmetric mean distance between the predicted and the true edge sets and by the relative error of the number of edge pixels, a proxy for the handover rate along a route,

$$\begin{aligned} \Delta_{\mathcal{E}} &= \frac{1}{2}\Big[\frac{1}{|\widehat{\mathcal{E}}|}\sum_{e\in\widehat{\mathcal{E}}}\min_{e'\in\mathcal{E}}\|e-e'\| + \frac{1}{|\mathcal{E}|}\sum_{e\in\mathcal{E}}\min_{e'\in\widehat{\mathcal{E}}}\|e-e'\|\Big],\\ \rho_{\mathcal{E}} &= \frac{|\widehat{\mathcal{E}}| - |\mathcal{E}|}{|\mathcal{E}|}. \end{aligned} \tag{35}$$

Selecting K of the $|\mathcal{B}|$ sites to maximise the number of covered pixels is a monotone submodular maximisation, for which the greedy algorithm attains a $(1-1/e)$ approximation of the optimum of the function it is given [16]; site selection with propagation models is a classical planning task [15]. With $C_{140}(S)$ the share of pixels covered at 140 dB by a set S of sites, the greedy plan on a map is $S_0 = \varnothing$ and

$$S_k = S_{k-1}\cup\Big\{\arg\max_{b\in\mathcal{B}\setminus S_{k-1}} C_{140}(S_{k-1}\cup\{b\})\Big\}, \qquad k = 1,\dots,K, \tag{36}$$

run once on the *predicted* map, giving $\hat{S}_K$, and once on the *true* map, giving $S^\star_K$; both are scored on the true map, and the planning regret is

$$R_K = C^{\mathrm{true}}_{140}(S^\star_K) - C^{\mathrm{true}}_{140}(\hat{S}_K). \tag{37}$$

For dimensioning, the number of sites the greedy order on the predicted map needs to cover 90% of the pixels is also counted, together with the true coverage those sites achieve.

A planner must also decide which pixels to *promise* as covered. For a transfer model and a held-out city c, let

$$r_p = \min\{P_L(p, s(p)), L_{\max}\} - \widehat{P_L}(p, \hat{s}(p)) \tag{38}$$

be the true serving path loss of pixel p, censored at the cut-off of the corpus, minus the predicted one, and q_c the $(1-\alpha)$ quantile of $\{r_p\}$ over the pixels of c. The censoring keeps q_c finite in a city where more than a fraction α of the pixels is reached by no site; where the model itself predicts no reaching site, $r_p = -\infty$ and the pixel is never claimed.

Proposition 4 (City-level coverage certificate)**.** *Let the n cities be exchangeable and the training procedure symmetric in its source cities. With q_c computed from the leave-one-city-out model of each city c, let $\hat{q}_t$ be the j-th largest of $\{q_c\}_{c\neq t}$ for a target city t, and claim pixel p as covered at level $L < L_{\max}$ when $\widehat{P_L}(p,\hat{s}(p)) + \hat{q}_t \le L$. Then, with probability at least $(n-j)/n$ over the draw of the target city, at most a fraction α of its pixels are claimed but not covered.*

Proof. Each q_c is a function of city c and of the multiset of the other cities, through the same symmetric procedure, so exchangeable cities give an exchangeable vector $(q_1, \dots, q_n)$. The event that q_t exceeds the j-th largest of the other values means that q_t is among the j largest of the vector, with no tie at the j-th place; by exchangeability its probability is at most j/n, hence $\Pr(q_t \le \hat{q}_t) \ge (n - j)/n$. On this event, a claimed pixel with $r_p \le \hat{q}_t$ has a finite prediction and satisfies $\min\{P_L(p, s(p)), L_{\max}\} = \widehat{P_L}(p, \hat{s}(p)) + r_p \le \widehat{P_L}(p, \hat{s}(p)) + \hat{q}_t \le L < L_{\max}$, hence $P_L(p, s(p)) \le L$ and the pixel is covered. Claimed but uncovered pixels therefore have $r_p > \hat{q}_t \ge q_t$, and by the definition of the quantile at most a fraction α of the pixels satisfy this. □

The guarantee is at the level of cities, the unit of shift, in the spirit of conformal prediction [33, 34]; it needs no model of the residuals, and the nested selection of Section 4.3 provides the required symmetry. The rank j trades confidence for coverage: $j = 1$ (the largest margin of the other cities) gives probability $(n - 1)/n$, $j = 2$ gives $(n - 2)/n$ with a smaller margin. The certificate is reported at $L = 140\,\text{dB}$ for $\alpha \in \{0.05, 0.10\}$ and $j \in \{1, 2\}$ (the results at 130 dB are released with the code) with the claimed and the falsely claimed share of the pixels, and compared with the naive claim $\widehat{P_L}(p, \hat{s}(p)) \le L$.

4.7. Robustness analyses and inference

Six analyses test whether the conclusions depend on the choices of the main corpus, and one protocol governs every comparison. *Mixed-path composition.* The framework of [11], following MAPLE [10], composes the loss of a link from every tile its straight path crosses. The horizontal segment from each site to each pixel is split at the tile borders (sampled every 10 m), so that tile t holds a length L_t of it, between the distances d_t^{in} and d_t^{out} from the site, and two readings of the composition are evaluated as a propagation law. The *piecewise* reading is a multi-slope law along the path and the *path-averaged* reading averages the parameters along the path,

$$\begin{aligned} P_{L,\text{pw}} &= A_s + \sum_t 10\gamma_t \log_{10} \frac{\max(d_t^{\text{out}}, d_0)}{\max(d_t^{\text{in}}, d_0)}, \\ P_{L,\text{avg}} &= \sum_t \frac{L_t}{d_h}[A_t + \gamma_t x], \end{aligned} \tag{39}$$

with the intercept A_s of the site's tile, the exponent of the tile each stretch of distance lies in, and the horizontal length d_h. Both reduce to Eq. (4) when the whole path lies in one tile. A third, literal reading that sums one log-distance loss per crossed tile is reported in Section 5.6. All are linear in the tile parameters, which are fitted jointly to all links of a city by least squares with a weak ridge penalty ($\lambda = 10$) towards the best city-wide pair; they are evaluated with tiles of 100 m and, as in [11], of 200 m, 500 m and 1000 m, always on the pixels of the main corpus. The labels of the tiles crossed by at least 30 links are transferred with the source median, kernel transfer and GBDT under the nested protocol (one seed), and the predicted maps are scored as above. *Network size.* Every zero-shot predictor is also scored on the network of the first ten sites selected by the siting rule (one site per 0.9 km^2, inter-site distance about 1020 m). *Siting and materials.* The models trained on the main corpus are applied unchanged to the three replicates; the tile model is scored with labels fitted on the replicate itself and with the labels of the same tiles from the main corpus. *Tile size.* Labels, descriptors, zero-shot transfer (median, ridge, GBDT) and the network evaluation are repeated for $G_s \in \{50, 100, 200\}$ m. *Censoring.* Links beyond the cut-off are missing, which can bias a least-squares slope. The labels are refitted with the cut lowered to 140 and 130 dB and, for 300 random tiles per city, the least-squares slope is compared with that of a censored-Gaussian (Tobit) model [35] that also uses the (pixel, site) pairs beyond the cut. *Second band.* The tile and link pipelines are repeated at 7.5 GHz, and the 3.5 GHz link models are applied to the second band with the free-space offset $20\log_{10}(f_2/f_1)$ of Eq. (8), their knife-edge feature evaluated at the new wavelength. *External scenes.* The Munich, Étoile and Florence scenes distributed with Sionna RT were modelled independently of the corpus of this study, with detailed meshes and several ITU materials (brick, concrete, marble, metal, wood). Building footprints and heights are derived from their meshes by vertical ray casting on a 2 m grid, 8 rooftop sites per scene are placed with the siting rule of the main corpus (minimum separation 150 m, the scenes being smaller), the scenes are ray traced with the corpus settings, and the following predictors are evaluated without any retraining: the tile labels, the source median, ridge and GBDT fitted on all eleven cities under the nested rule, and the mean of the eleven leave-one-city-out link models, zero-shot and with the per-site calibration of Section 4.4 from $k \in \{10, 25\}$ surveyed tiles (ten random draws, and the two designs), scored on the unsurveyed pixels. The San Francisco scene has terrain and would require a terrain-following measurement surface; it is not used.

Every city is the target once (leave-one-city-out) and the other cities are training cities, inside which the nested selection of Section 4.3 takes place. Tiles of one city are spatially correlated, so tile-level tests would overstate significance; the unit of inference is the target city. For a comparison of two predictors with per-city differences $\Delta_1, \dots, \Delta_n$ of a metric, the mean difference $\bar{\Delta}$ is reported with a 95% percentile bootstrap interval over cities [53] from $B = 10{,}000$ resamples,

$$\text{CI}_{95} = [\bar{\Delta}^*_{(0.025B)},\ \bar{\Delta}^*_{(0.975B)}], \tag{40}$$

the exact two-sided Wilcoxon signed-rank p-value [54] and the number of cities in which the first predictor is better. Within each family of m comparisons the p-values are ordered, $p_{(1)} \le \cdots \le p_{(m)}$, and Holm's step-down correction [55] rejects the hypotheses in that order while

$$p_{(i)} \le \frac{0.05}{m - i + 1}, \tag{41}$$

stopping at the first failure. Standard deviations across cities use $n - 1$. With 11 cities an exact Wilcoxon test can reach $p < 0.05$ only when nearly every city agrees; the power limit is stated where it matters.

5. Performance evaluation

This section reports the experimental setting and the results of the evaluation described in Section 4. The results are presented in the order of the pipeline of Fig. 2: the corpora and the precision of the tile labels, the transfer of tile parameters, the network-level evaluation and its decomposition, the mixed-path compositions, survey design and certified coverage, and the robustness analyses. Every table and figure is generated from the released result files, and every comparison is accompanied by the city-level statistics of Section 4.7, which are collected in Section 5.9.

5.1. Experimental setting

The setting is fixed by the configuration files of the released pipeline, so that a reader can regenerate every result under the same conditions. All corpora were generated and all analyses were executed on a virtual machine with four cores of an Intel Xeon processor at 2.1 GHz and 15 GB of memory under Ubuntu 24.04, without a GPU. Ray tracing uses Sionna RT 2.1.0 on the LLVM backend of Mitsuba 3.9; the analysis uses Python 3.13 with scikit-learn 1.9, LightGBM 4.7 and PyTorch 2.14. Table 3 lists the parameters of the corpus generation, the models and the evaluation. The ray tracing of the main corpus took 4.7 h on the CPU (LLVM) backend (77 s per site); the analysis that follows it, from the tile labels to the certificates, took 8.0 h in total on 4 cores (Table 4), of which the heaviest step, `18_mixed_path.py`, accounts for 4.1 h.

The evaluation metrics are those of Section 4.6, and are summarised as follows.

1. **Parameter error:** root-mean-square error (RMSE) of the transferred exponent γ and intercept P_{L_0} over the tiles of the held-out city.
2. **Path-loss error:** RMSE and bias over the ray-traced links, and the censored RMSE of Eq. (32) over all (pixel, site) pairs.
3. **Link existence:** precision and recall of the predicted links below the cut-off, and the share of pixels whose reached state is predicted correctly.
4. **Serving site:** share of the reached pixels assigned to the correct serving site, Eq. (33).
5. **SINR and coverage:** mean absolute SINR error, Kolmogorov–Smirnov distance between the SINR distributions, coverage agreement and coverage-share error at 130 dB, and the error of the mean spectral efficiency, Eq. (34).
6. **Cell edges:** displacement of the predicted cell edges and relative error of their total length, Eq. (35).
7. **Planning regret and dimensioning:** coverage lost by selecting K sites on the predicted map instead of the true map, Eq. (37), and the number of sites the predicted map claims to need for 90% coverage together with their true coverage.
8. **Certified coverage:** share of the pixels claimed covered under the certificate of Proposition 4 and share claimed but not covered.

Table 3: Parameters of the corpus generation, the models and the evaluation.

Parameter	Value
Scene size; building source	3×3 km; Overture Maps release 2026-09-23.0
Materials	ITU concrete (buildings), medium dry ground
Sites per city; mast above roof; minimum separation	20; 6 m; 250 m
Carrier frequencies	3.5 and 7.5 GHz
Receiver grid; receiver height	10 m cells; 1.5 m
Ray budget per site; interaction depth	1×10^8; 5
Path-loss cut-off $L_{\max}$	150 dB
Tile side G_s; reference distance d_0	100 m; 100 m
Minimum links per labelled tile	30
Descriptors per tile; link features	32; 46
Transmit power; bandwidth; noise figure	46 dBm; 20 MHz; 7 dB
Coverage levels	120, 130 and 140 dB
Site planning	greedy, $K = 5$; dimensioning target 90%
Seeds; inner folds of the expensive operators	5; 3
Surveyed tiles k: tile correction; site calibration	3–100; 10, 25, 50
Fixed prior strength κ (calibration ablation)	50 links
Certificate	$\alpha \in \{0.05, 0.10\}$; $L = 140$ dB; rank $j \in \{1, 2\}$
Bootstrap resamples; significance level	10,000; 5% (Holm)

Table 4: Wall-clock time of the analysis steps of `make reproduce` on the machine of Section 5.1 (4 cores, x86_64). The ray tracing of the corpora is separate (4.7 h for the main corpus).

Script	runs	minutes
`02_build_tiles.py`	2	3
`03_make_splits.py`	2	0
`04_fit_baselines.py`	2	43
`06_train_neural.py`	1	8
`05_kshot.py`	1	0
`07_label_noise.py`	1	1
`08_degeneracy.py`	1	0
`11_link_transfer.py`	2	123
`09_network_eval.py`	2	3
`19_tile_variants.py`	1	15
`13_certified_coverage.py`	1	0
`12_tile_size.py`	1	12
`14_censoring.py`	1	1
`15_siting.py`	1	11
`17_external_scenes.py`	1	0
`18_mixed_path.py`	4	247
`16_band.py`	1	14
`10_stats.py`	1	0
Total	26	482

5.2. Corpora, solver convergence and tile labels

Before any model is compared, two properties of the data fix what a model can be expected to achieve: the Monte-Carlo noise of the ray-traced truth and the precision of the tile labels fitted from it. Both are quantified here, together with the descriptive statistics of the corpora.

The corpus contains 11 cities, 105,800 buildings, 220 sites and 8,977,400 ray-traced links (Table 5); 10,428 tiles have at

least 30 links and are labelled. Figure 4 shows one city. The share of buildings whose height is imputed from the city median ranges from 2% (New York) to 50% (London). The radio-map solver is a Monte-Carlo method: a cell that no ray reaches reports zero power, so the set of reached cells and their values depend on the ray budget. Figure 5 compares budgets with a reference run of 1×10^9 rays for 6 sites in 3 cities. At the 1×10^8 rays per site used for the corpus, path loss below 120 dB agrees with the reference to at most 0.80 dB RMS and path loss below 140 dB to 1.4–2.7 dB, with a mean difference of at most 0.09 dB; the number of cells reached below 140 dB is at least 89.2% of the reference, and below the cut-off at least 85.7%. These values are the noise floor of the truth: no model can be expected to match the corpus more closely than 1.4 dB to 2.7 dB near the cut-off, and up to 3.8 dB for the faintest links.

Table 6 summarises the labels. City medians of the tile exponent range from 2.39 (Vienna) to 3.15 (Hamburg); their standard deviation across cities, 0.27, is 0.3 times the typical standard deviation of the tiles within a city (0.60–0.96), so most of the variation of the tile exponent lies within cities rather than between them (Fig. 6a). The fit residual of the log-distance law inside a tile is 11.2–12.7 dB. The city-wide exponent of a single law over all links is below the free-space value in 4 cities (Chicago: 0.81), the joint effect of high sites and of the censoring of far links (Section 5.8); the tile labels, fitted inside 100 m, do not show this artefact. Without shrinkage, the OLS exponents span −4.8 to 8.8, the ill-conditioning of Section 4.2; the empirical-Bayes estimator of Eq. (10) (median weight 0.36) narrows this to −0.90 to 6.45. The descriptors carry signal but no single one explains much of the variation (Fig. 6b): the strongest correlation with the exponent is $r = -0.45$, for footprint density.

Label precision is the decisive quantity for everything that follows (Fig. 7). The standard error of the exponent under independent residuals is 0.141 (RMS over tiles); the cluster-robust error of Eq. (12) with sites as clusters is 0.456, 3.2 times larger, and the split-half estimate of Eq. (13) extrapolated to the full data is 0.686. For the intercept the corresponding errors are 1.49 and 5.40 dB. The exponent of a tile therefore depends strongly on which sites serve it, and a label error of this size is comparable to the differences between cities (Table 6).

5.3. Transfer of tile parameters

This subsection answers the first part of the transfer question: how well the tile parameters of a city that took no part in training can be predicted from its descriptors, zero-shot and with a small survey. The operators of Section 4.3 are compared under the nested protocol of Fig. 3, the kernel operator is placed relative to Proposition 2, and the few-shot corrections of Section 4.4 are evaluated against the number of surveyed tiles.

Table 7 and Fig. 8 report the exponent error on every held-out city. The source median attains 0.871; kernel transfer 0.801; the Gaussian process with learned length scales 0.762; the learned regressors 0.765 (ridge), 0.768 (MLP), 0.766 (neural encoder) and 0.764 (GBDT). The domain-adaptation variants do not improve on their base learners (CORAL 0.789 against ridge 0.765; GBDT-IW 0.772 against GBDT 0.764), and the standard models are worse than the median (3GPP UMa 1.003 for line of sight and 1.453 for non-line of sight, COST-231 Hata 1.057). The lowest mean error is that of the GP-ARD, but the learned operators differ by less than their spread across cities. For the intercept at 100 m the learned operators reduce the error from 13.8 dB (median) to 9.5 dB (GBDT). The hyper-parameters selected by the inner loop are listed in Table 8. At the level of cities (Table 27), GBDT changes the error against the median by −0.106 (95% CI [−0.157, −0.052], $p = 0.010$, better in 9 of 11 cities) and against kernel transfer by −0.037 ([−0.061, −0.016]); kernel transfer against the median gives −0.070 ([−0.129, −0.001], $p = 0.083$), whereas learning the metric of the same kernel gives −0.039 ([−0.063, −0.021], $p = 0.002$, better in 10 of 11 cities), significant after Holm's correction. Over the 12 comparisons with the source median, none of the improvements survive Holm's correction at eleven cities, although the learned operators are better than the median in 8 to 9 of 11 cities; the significant differences in the other direction are 3GPP UMa NLoS. The hardest target is Chicago, where the best operator (Source median) reaches 0.834.

Table 9 places the tuned kernel operator relative to Proposition 2. The selected bandwidths are $\sigma = 0.1$–0.2 and the median ϵ_u ranges from 73.6 to 306: 0.0% of the target tiles satisfy the condition of part (a), and the largest single weight has median 0.00, so the operator is not a nearest-neighbour copy either. Where the largest bandwidth is selected (9 of 11 targets), it averages over up to 29% of the source tiles and its predictions spread over as little as 0.32 of the spread of the target labels: it approaches the source mean although part (a) does not hold. Across the targets the prediction spread is 0.32–0.76 of that of the labels. The disadvantage of the operator is the noise-transmission term of Eq. (7) combined with an isotropic distance in which every descriptor counts equally; the Gaussian process of Eq. (16), which learns one length scale per descriptor, is the kernel operator that removes most of the gap to the learned regressors.

Surveying target tiles helps the tile exponent little at small k (Table 10, Fig. 9). The plain mean-residual correction is worse than the zero-shot model at $k = 3, 5, 10$ (GBDT at $k = 3$: 0.849 against 0.764 zero-shot): the labels of the anchors carry the site-dependent error measured above, and a correction fitted to a few of them adds more variance than it removes. The empirical-Bayes correction of Eq. (22), which weights the anchors by their precision and shrinks the correction when the inner cities show no systematic bias, is worse than the zero-shot model at no value of k (GBDT at $k = 3$: 0.761; at $k = 10$: 0.759, −0.018 against the plain correction, [−0.028, −0.006]), but its gain is small: with ten random tiles it improves the GBDT error in 9 of 11 cities, by 0.6% on average (−0.005, [−0.011, −0.001]), and with $k = 100$ tiles by 1.9% (0.750). The anchors measure the site-dependent label, not an environmental constant, so even many of them move the transferred exponent little. The precision-weighted residual model of Eq. (24) gives 0.775 at $k = 10$. Contiguous survey areas are worse than random tiles (GBDT at $k = 10$: 0.765), because neighbouring tiles repeat the same information, and the facility-location design of Eq. (25),

Table 5: Corpus. Height source: share of buildings with an explicit height attribute, with a floor count, or imputed from the city median. Site height: antenna height above ground of the 20 sites. Links: ray-traced (pixel, site) pairs below the 150 dB cut-off. Tiles: labelled tiles (≥ 30 links).

		Height source (%)			Site height (m)					
City	Buildings	tag	storeys	imputed	Median (m)	min	med.	max	Links	Tiles
Amsterdam	21,135	83	2	15	15.5	39	47	67	736,255	948
Barcelona	12,337	12	64	23	19.2	44	56	91	609,050	942
Berlin	6,036	15	63	22	16.0	44	75	374	1,009,619	957
Chicago	2,952	70	4	25	15.9	175	200	346	1,069,552	961
Hamburg	4,335	17	42	41	16.0	38	51	116	1,152,125	961
London	11,891	8	42	50	12.8	48	64	161	931,288	940
Munich	8,616	9	49	42	12.8	32	38	97	794,056	950
New York	7,126	98	1	2	18.6	232	318	478	827,510	960
Paris	14,383	3	64	33	19.2	38	46	96	529,649	910
Prague	9,907	1	70	29	16.0	38	41	222	783,805	960
Vienna	7,082	26	60	14	16.0	41	45	84	534,491	939
Total	105,800								8,977,400	10,428

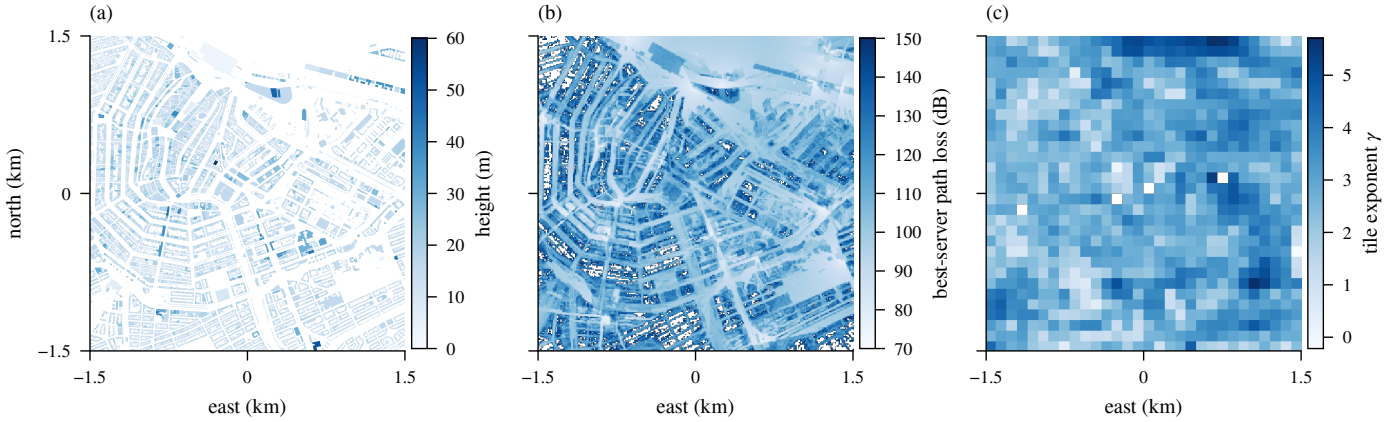


Figure 4: Amsterdam. (a) Building heights of the scene. (b) Ray-traced path loss to the best site. (c) Tile exponents $\hat{\gamma}_t$.

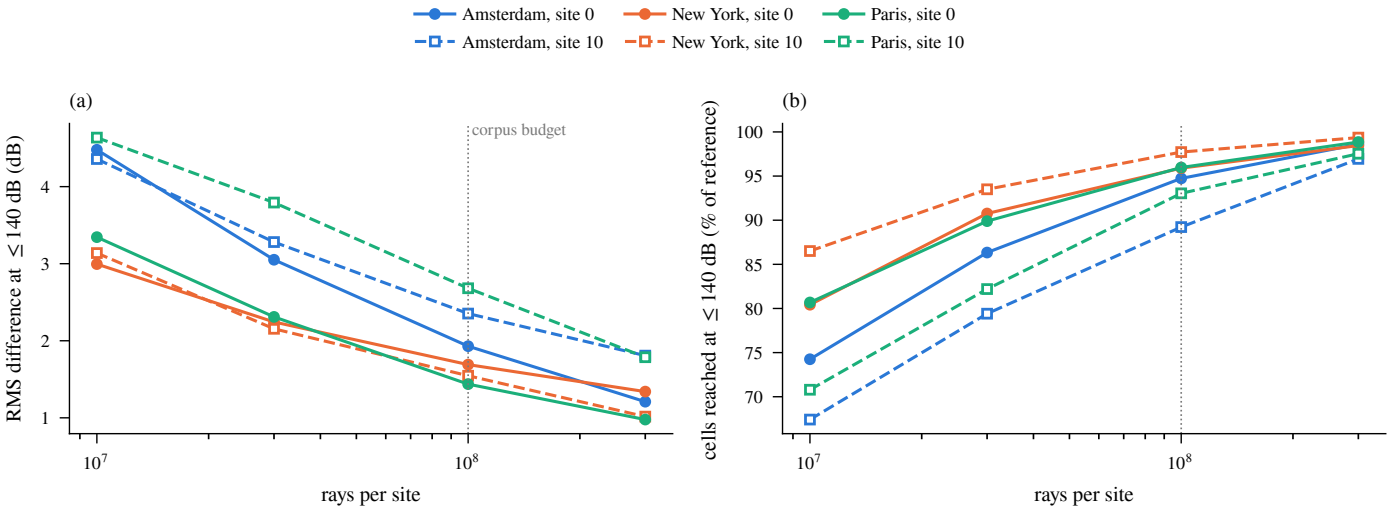


Figure 5: Ray-budget convergence of the radio-map solver for 6 sites in 3 cities, relative to a reference run of 1×10^9 rays. (a) RMS path-loss difference to the reference over the cells reached by both below 140 dB. (b) Cells reached below 140 dB as a share of the reference (Monte-Carlo noise can exceed 100%). The dotted line marks the corpus budget of 1×10^8 rays.

which spreads the survey over the descriptor space without any measurement, gives 0.755 at $k = 10$ (−0.004 against random

Table 6: Tile labels per city. City-wide fit: OLS over all links. Tile labels: median and standard deviation of $\hat{\gamma}_t$ and median $\hat{P}_{L_0,t}$ (dB, at $d_0 = 100$ m); $\hat{\tau}$: between-tile standard deviation of the slope; w_t: shrinkage weight; RMSE: link-weighted fit residual (dB).

	City-wide fit		Tile labels					
City	γ	P_{L_0}	med. γ	s.d. γ	med. P_{L_0}	$\hat{\tau}$	med. w_t	RMSE
Amsterdam	2.47	100.5	2.84	0.93	100.9	1.06	0.27	11.3
Barcelona	1.92	105.4	2.43	0.96	105.5	1.14	0.28	11.6
Berlin	2.65	98.2	2.82	0.81	97.5	1.00	0.36	11.6
Chicago	0.81	115.6	2.79	0.83	97.2	1.19	0.54	12.7
Hamburg	2.14	100.2	3.15	0.72	92.2	0.94	0.37	11.4
London	1.72	104.4	2.40	0.60	103.3	0.79	0.42	11.2
Munich	2.12	104.0	3.00	0.80	99.8	0.92	0.33	11.2
New York	1.11	115.9	2.69	0.79	102.2	1.18	0.53	12.6
Paris	2.06	103.6	2.41	0.95	106.7	1.13	0.29	11.8
Prague	2.48	100.9	2.92	0.77	99.0	0.95	0.37	11.4
Vienna	2.29	106.1	2.39	0.92	108.4	1.09	0.32	11.4

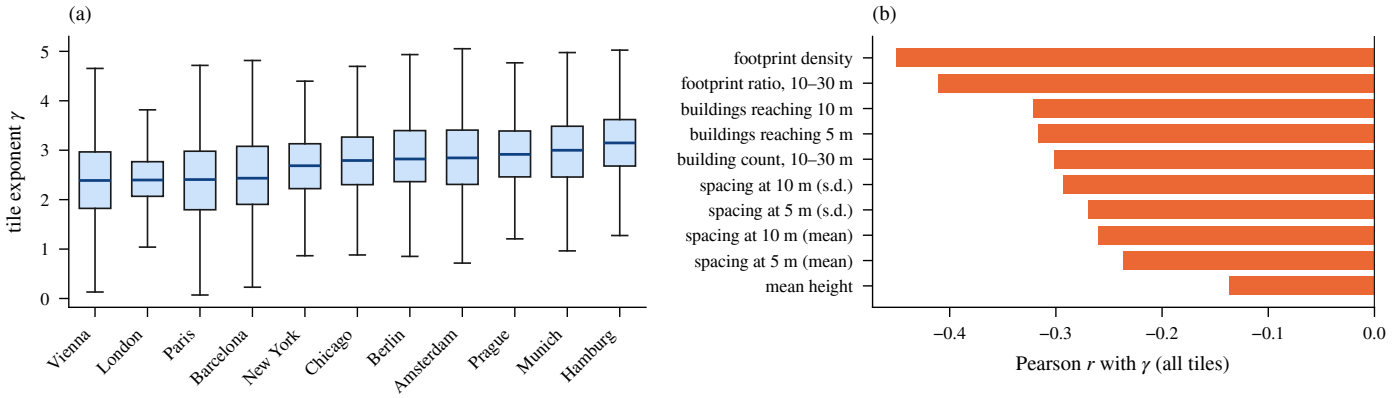


Figure 6: (a) Tile exponents per city (box: quartiles; whiskers: 1.5 times the interquartile range). (b) The ten descriptors most correlated with the tile exponent over all 10,428 tiles.

Table 7: Zero-shot exponent RMSE per held-out city (mean of five seeds for stochastic operators; best per row in bold). Last row: mean P_{L_0} RMSE (dB).

Target city	Median	UMa-L	UMa-N	COST	Kernel	GP	k-NN	Ridge	CORAL	MLP	Enc.	GBDT	IW
Amsterdam	0.944	1.146	1.394	1.042	0.818	0.784	0.793	0.780	0.817	**0.776**	0.782	0.786	0.792
Barcelona	0.999	0.998	1.730	1.303	0.811	**0.780**	0.800	0.784	0.823	0.786	0.786	0.788	0.787
Berlin	0.828	1.052	1.319	0.955	0.792	**0.749**	0.771	0.758	0.862	0.755	0.751	0.751	0.756
Chicago	**0.834**	1.006	1.414	1.030	0.935	0.893	0.960	0.879	0.875	0.892	0.894	0.885	0.894
Hamburg	0.844	1.162	1.078	0.761	0.645	0.641	0.627	0.632	0.637	0.629	**0.620**	0.645	0.651
London	0.687	0.639	1.599	1.110	0.641	0.642	0.650	**0.635**	0.636	0.650	0.643	0.661	0.673
Munich	0.854	1.120	1.229	0.881	0.767	**0.756**	0.766	0.765	0.793	0.764	0.768	0.758	0.758
New York	**0.793**	0.907	1.489	1.073	0.965	0.824	0.825	0.868	0.830	0.864	0.839	0.827	0.863
Paris	0.997	0.977	1.748	1.308	0.857	**0.801**	0.826	0.806	0.833	0.813	0.808	0.803	0.808
Prague	0.817	1.086	1.219	0.858	0.706	0.678	0.679	0.673	0.753	**0.671**	0.681	0.672	0.680
Vienna	0.978	0.938	1.761	1.311	0.873	0.836	0.848	0.836	**0.819**	0.843	0.852	0.829	0.829
Mean	0.871	1.003	1.453	1.057	0.801	**0.762**	0.777	0.765	0.789	0.768	0.766	0.764	0.772
S.d.	0.098	0.147	0.234	0.190	0.107	0.081	0.096	0.086	0.082	0.087	0.087	0.077	0.079
P_{L_0} (dB)	13.8	21.6	13.8	17.9	10.2	9.5	9.7	9.6	10.3	9.5	9.6	9.5	9.5

tiles, [−0.010, +0.001], better in 8 of 11 cities).

5.4. Network-level evaluation

The parameter errors of the previous subsection matter only through the quantities a planner derives from the map. This subsection scores every predictor over the complete receiver grid of the labelled tiles, through path loss, link existence, serving site, SINR, coverage, cell edges, dimensioning and the regret of a site plan (Section 4.6).

Table 11 and Fig. 10 give the means over the held-out cities

Table 8: Hyper-parameters selected by the inner leave-one-training-city-out loop of each split. N: training tiles; σ: kernel bandwidth; k: neighbours; epochs and rounds: median best iteration over the inner folds (GBDT: for γ); link model: rounds of the regressor and of the hurdle classifier.

					MLP		GBDT	Link model	
Target	N	σ	k	α_{ridge}	α	epochs	rounds	rounds	rounds (clf.)
Amsterdam	9,480	0.2	200	1	0.01	8	58	71	79
Barcelona	9,486	0.2	200	1	0.01	4	78	87	85
Berlin	9,471	0.2	200	1	0.0001	9	83	83	55
Chicago	9,467	0.1	200	10	0.01	8	67	51	91
Hamburg	9,467	0.2	200	1	0.01	15	72	60	75
London	9,488	0.2	200	1	0.01	7	70	98	78
Munich	9,478	0.2	200	1	0.0001	5	70	85	82
New York	9,468	0.1	200	10	0.0001	14	76	56	110
Paris	9,518	0.2	200	1	0.01	8	72	56	69
Prague	9,468	0.2	200	1	0.01	10	86	85	95
Vienna	9,489	0.2	200	1	0.01	4	69	139	105

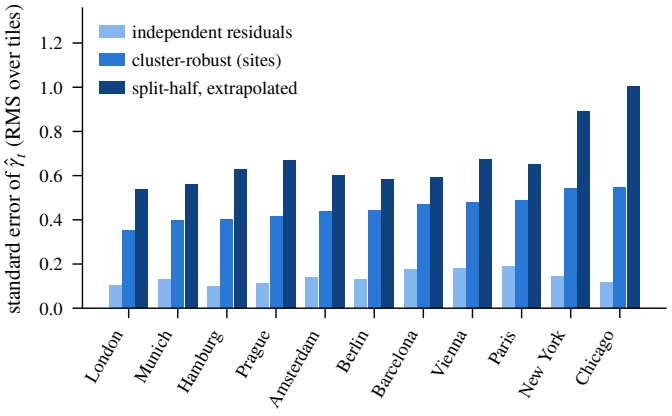


Figure 7: Standard error of the tile exponent per city (RMS over tiles) under independent residuals, the cluster-robust estimator with sites as clusters, Eq. (12), and the split-half estimator extrapolated to all sites, Eq. (13).

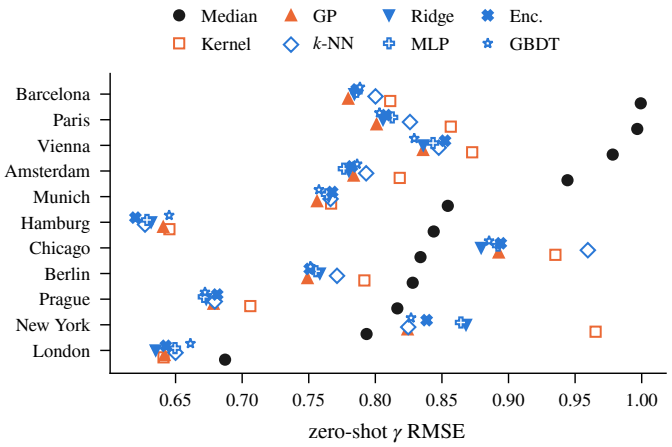


Figure 8: Zero-shot exponent RMSE per held-out city, ordered by the error of the source median.

and Fig. 11 the city-level values of the three reference predictors. The first row of Table 11 is the tile model with the true labels (oracle), which isolates the error of the tile model itself. The oracle has a path-loss error of 11.6 dB over the ray-traced

Table 9: Kernel transfer on each held-out city: selected bandwidth; median $\epsilon_u = D_u/2\sigma^2$ and share of tiles with $\epsilon_u < 1$ (Proposition 2a); median effective-neighbourhood fraction; median largest weight (Proposition 2b); standard deviation of the predictions over that of the labels.

Target	σ	med. ϵ_u	ϵ_u<1 (%)	n_{eff}/N (%)	med. $w_{\max}$	spread
Amsterdam	0.2	78.7	0	16.42	0.002	0.41
Barcelona	0.2	74.8	0	19.68	0.001	0.37
Berlin	0.2	73.6	0	26.67	0.001	0.34
Chicago	0.1	306	0	0.46	0.061	0.63
Hamburg	0.2	75.8	0	17.04	0.001	0.45
London	0.2	75.5	0	24.22	0.001	0.60
Munich	0.2	74.9	0	29.42	0.001	0.39
New York	0.1	276	0	0.03	0.463	0.76
Paris	0.2	75.9	0	24.48	0.001	0.33
Prague	0.2	74.7	0	28.83	0.001	0.35
Vienna	0.2	73.6	0	27.56	0.001	0.32

Table 10: Few-shot correction: exponent RMSE on the unsurveyed tiles, mean over cities (and ten anchor draws for random and contiguous anchors). Left: the empirical-Bayes correction with random anchors. Right: GBDT with the plain bias correction, the precision-weighted residual model, contiguous anchors and the facility-location design (all empirical Bayes unless stated).

	EB correction, random tiles				GBDT			
k	Kernel	Ridge	MLP	GBDT	plain bias	residual	contig.	design
0	0.801	0.765	0.768	0.764	0.764	0.764	0.764	0.764
3	0.793	0.761	0.764	0.761	0.849	0.844	0.764	0.765
5	0.795	0.763	0.765	0.763	0.824	0.818	0.765	0.762
10	0.788	0.760	0.762	0.759	0.778	0.775	0.765	0.755
25	0.781	0.754	0.756	0.754	0.758	0.752	0.771	0.751
50	0.780	0.752	0.754	0.751	0.753	0.744	0.781	0.749
100	0.777	0.751	0.752	0.750	0.750	0.739	0.783	0.753

links and 13.9 dB over all pairs clipped at the cut-off, selects the correct serving site for 50.1% of the pixels (the nearest-site rule of Section 3.2) and misjudges the SINR by 7.9 dB on average. Transferred tile parameters reduce the path-loss error of the source median from 14.5 dB to 12.8 dB (GBDT; −1.75 dB, better in 11 of 11 cities) but remain above the oracle, and a plan of $K = 5$ sites made on their map loses 18.4% of the pixels against 9.2% for the oracle.

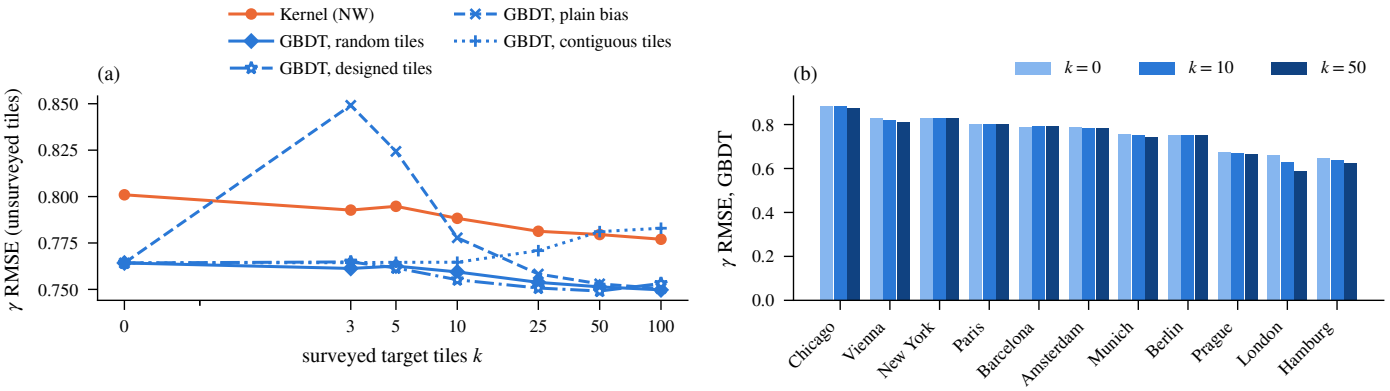


Figure 9: (a) Exponent RMSE on the unsurveyed tiles against the number of surveyed target tiles, for kernel transfer and for GBDT with the empirical-Bayes correction from random, designed and contiguous tiles and with the plain bias correction; ridge and the MLP lie within the line width of GBDT and are listed in Table 10. (b) GBDT per held-out city at $k \in \{0, 10, 50\}$.

Site-aware link transfer removes the ceiling of the tile model. Its path-loss error is 9.3 dB (8.8 to 9.6 dB across cities, Fig. 11a), 3.4 dB below tile transfer with GBDT (95% CI [−3.83, −3.10], better in 11 of 11 cities) and 2.3 dB below the tile model with the *true* labels ([−2.63, −2.08], 11 of 11). It selects the correct serving site for 68.4% of the pixels (+18.3 percentage points against GBDT, in 11 of 11 cities), misjudges the SINR by 5.7 dB on average and loses 0.7% of the pixels when planning $K = 5$ sites, against 18.4% for GBDT tile transfer and 9.2% for the true tile labels (Fig. 11c). Its hurdle decides the reach of every site: it predicts the reached set of pixels correctly for 95.8% of them, with a link precision of 81.7% and a recall of 79.2% (Table 12); the regressor alone, which keeps every link, has the same path-loss error over the ray-traced links but a censored error of 11.3 dB and a planning regret of 4.5%, because it lets distant, obstructed sites cover pixels they do not reach. The model relies on the propagation features, which account for 98% of the gain of its trees (Table 13); the single most important feature is the knife-edge loss (22%), and the three most important together account for 59%.

Table 14 turns the maps into mobility and dimensioning quantities. Every tile model draws the nearest-site cell pattern of Section 3.2: its cell edges lie 81 m from the true edges on average (76 m with the true labels), and their total length is in error by −92%, because the ray-traced serving pattern is fragmented by buildings and a route crosses many more cell edges than the nearest-site pattern suggests. Link transfer places the edges 7 m from the truth with an edge-length error of −8%. For dimensioning, the truth needs on average 6.1 of the 20 sites to cover 90% of the pixels at 140 dB. The tile models, which cannot represent the obstruction of distant sites, predict that a single site suffices (GBDT; 1.1 sites with the true labels, averages over the cities), and the sites they choose cover only 47.9% of the pixels in truth. Link transfer chooses 3.6 sites on average, which cover 80.4%; it still under-dimensions the network, but by far less (the regressor alone: a single site, 54.9%).

5.5. *Decomposition of the gap and physics baselines*

The gap between the tile model and link transfer measured above has three candidate causes: the absence of a reachability model, the absence of a line-of-sight state, and the per-tile summary of the link geometry itself. This subsection separates them with the receiver-tile variants of Section 4.7 and compares the learned link model with the non-learned per-link baselines of Section 4.5, which use the same geometry without training.

Table 15 and Fig. 12 attribute the gap. Giving GBDT tile transfer the hurdle of the link model changes its path-loss error from 12.8 dB to 12.8 dB over the ray-traced links, but its censored error from 14.5 dB to 11.2 dB and its planning regret from 18.4% to 0.9% (−17.5 percentage points, [−23.3, −12.4]): the cut-off, not the path-loss accuracy, is what makes the plans of the tile models unreliable. The dual-slope label of Eq. (11), which gives every tile a line-of-sight and an obstructed pair (Table 16: 6–25% of the links are in line of sight; median exponents 2.50 and 2.52), lifts the nearest-site restriction and changes the serving-site accuracy of GBDT from 50.1% to 62.0% (+11.8 points, [+9.2, +15.2]) and the path-loss error to 11.0 dB; with both, the tile model reaches 11.0 dB, 62.2% and a regret of 0.9%. With exact labels the same two additions bring the tile model to 10.0 dB, 61.8% and 0.7%. Link transfer remains better than the best tile variant with transferred labels (GBDT, dual slope in path loss) in 11 of 11 cities in path loss, in 11 in serving site and in 3 in regret (−1.65 dB, [−1.81, −1.50]), and than the dual-slope hurdle model with *exact* labels by −0.70 dB ([−0.81, −0.60]) and +6.6 points of serving site. What remains after the cut-off and the line-of-sight state is the geometry of the individual link: how far above the line of sight the obstruction rises, how many buildings the path crosses and how long it runs over them, which no per-tile summary carries.

The non-learned per-link baselines use exactly that geometry. Free space with a single knife edge, Eq. (28), gives 14.7 dB, the Deygout construction of Eq. (30) 21.7 dB and 3GPP UMa with the geometric line-of-sight state, Eq. (31), 17.9 dB; the best of them in path loss is Free space + knife edge (14.7 dB), in serving site 3GPP UMa, geometric LoS (62.3%) and in planning regret Free space + Deygout (0.1%). Link transfer is better than

Table 11: Network-level evaluation on the held-out cities (mean over cities). Oracle: tile model with the true labels. Path loss: RMSE and bias over the ray-traced links, and RMSE over all pairs with prediction and truth clipped at the cut-off (cens.). Serving site: share of reached pixels with the correct serving site. Coverage: agreement of covered pixels (serving path loss ≤ 130 dB) and error of the covered share. SINR: mean absolute error and Kolmogorov–Smirnov distance of the distributions. SE: error of mean spectral efficiency (b/s/Hz). Regret: coverage lost by planning $K = 5$ sites on the predicted map instead of the true map. True covered share: 85.3%; true mean spectral efficiency: 1.39 b/s/Hz. Few-shot rows are scored on the unsurveyed pixels.

	Path loss (dB)			Serving	Coverage 130 dB (%)		SINR		SE	Regret
Predictor	RMSE	bias	cens.	site (%)	agree	Δ	MAE (dB)	KS	Δ	K=5 (%)
Tile labels (oracle)	11.64	+0.00	13.94	50.1	87.2	8.7	7.89	0.283	−0.599	9.2
Source median	14.51	+3.72	15.89	50.1	83.9	12.9	7.68	0.269	−0.606	29.8
3GPP UMa NLoS	22.75	+17.29	15.35	50.1	69.2	−14.3	7.56	0.134	−0.164	9.0
COST-231 Hata	25.38	+20.99	16.30	50.1	53.4	−38.1	7.67	0.204	−0.441	10.8
Kernel (NW)	12.91	+1.16	14.68	50.1	84.5	12.4	7.92	0.275	−0.615	24.3
GP-ARD	12.77	+0.97	14.45	50.1	84.9	11.2	7.92	0.274	−0.614	15.2
k-NN	12.80	+1.01	14.53	50.1	84.9	12.3	7.97	0.274	−0.611	32.5
Ridge	12.76	+1.10	14.45	50.1	84.9	10.4	7.96	0.280	−0.617	19.9
CORAL + ridge	13.03	+2.18	14.25	50.1	84.5	10.2	7.98	0.292	−0.641	18.4
MLP	12.75	+0.89	14.50	50.1	84.8	11.5	7.95	0.276	−0.613	24.2
Neural encoder	12.81	+0.98	14.59	50.1	84.9	12.0	7.96	0.278	−0.616	28.6
GBDT	12.75	+0.95	14.52	50.1	85.0	11.6	7.93	0.274	−0.614	18.4
GBDT-IW	12.78	+0.94	14.57	50.1	85.0	11.5	7.95	0.275	−0.613	15.5
GBDT + 10 tiles	12.73	+0.96	14.48	50.1	85.1	11.6	7.93	0.274	−0.612	19.0
GBDT + 10 designed tiles	12.74	+1.10	14.40	50.1	85.0	11.5	7.92	0.269	−0.600	16.5
Link transfer	9.32	+0.00	9.21	68.4	90.0	3.1	5.74	0.046	−0.030	0.7
Link, regressor only	9.32	+0.00	11.35	69.0	89.9	3.4	6.14	0.151	−0.291	4.5
Link + 25 tiles	9.35	+0.04	9.17	66.4	89.8	3.3	5.88	0.046	−0.020	0.7
Link + 25 tiles, stratified design	9.41	−0.42	9.23	65.3	89.5	4.0	6.04	0.053	−0.019	0.7

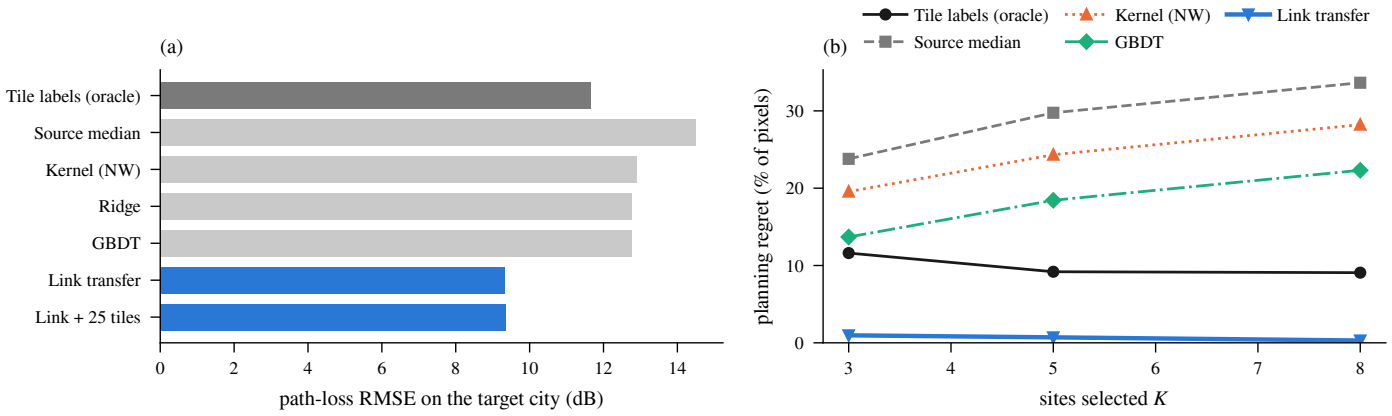


Figure 10: (a) Path-loss RMSE on the held-out cities. (b) Planning regret against the number of selected sites.

Table 12: Link existence. Reach: share of pixels whose reached/unreached state is predicted correctly; precision and recall of the predicted links below the cut-off; censored RMSE over all pairs. The tile models reach every pixel from every site.

Predictor	reach (%)	precision (%)	recall (%)	cens. RMSE (dB)
Tile labels (oracle)	95.2	45.5	100.0	13.94
GBDT	95.2	45.5	100.0	14.52
Link transfer	95.8	81.7	79.2	9.21
Link, regressor only	95.2	45.5	100.0	11.35

the best physics baseline in 11 of 11 cities in path loss, in 11 in serving site and in 3 in regret; against the Deygout model the differences are −12.34 dB ([−15.83, −9.53]), +12.6 points of serving site and +0.6 points of regret, and against UMa with geometric line of sight −8.56 dB ([−9.14, −7.94]) and +6.1 points. The planning regret is the one quantity in which a physics baseline is as good: the Deygout construction with the deterministic hurdle plans five sites with a regret of 0.1% against 0.7% for link transfer, although its path loss is in error by 21.7 dB, because a large but nearly uniform overestimate does not change which sites cover the most pixels. Of the 25 comparisons of link transfer with the tile variants and the physics baselines, 20 hold in every city and 20 survive Holm's correction. The learned link model thus adds accuracy in path loss, serving site, SINR and coverage to the diffraction physics it is built on; a per-link physics model with a reach model is already enough for a five-

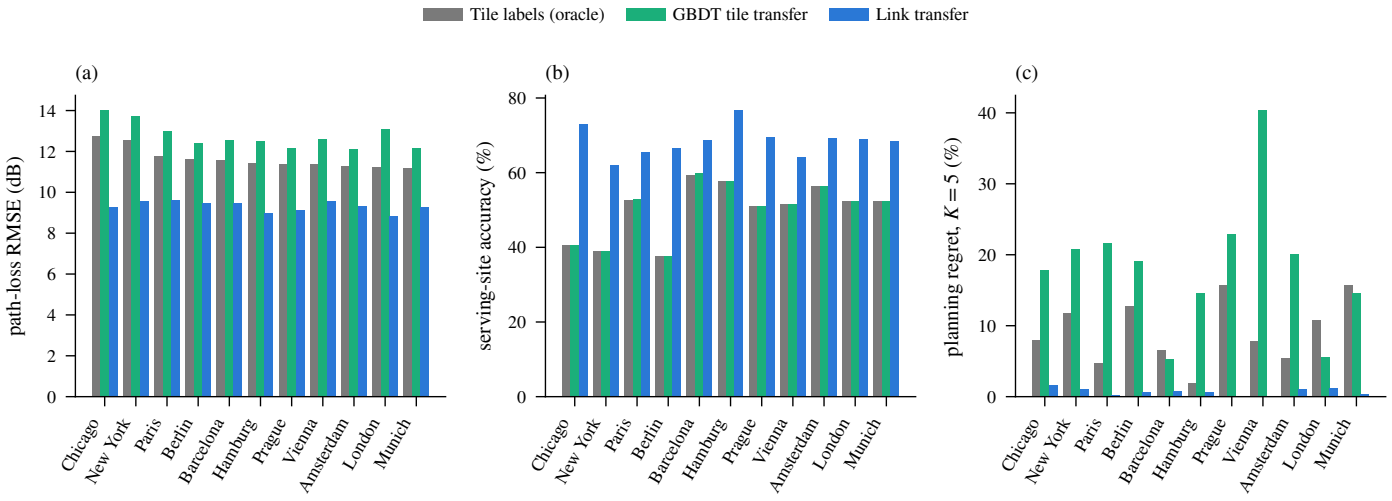


Figure 11: Per-city results of the tile model with exact labels, GBDT tile transfer and link transfer (mean of five seeds), ordered by the path-loss error of the exact labels. (a) Path-loss RMSE. (b) Serving-site accuracy. (c) Planning regret for $K = 5$.

Table 13: Link transfer: the ten features with the largest share of the total split gain (mean over folds and seeds).

Feature	gain share (%)
knife-edge loss	22.0
max. excess height	19.2
buildings crossed	17.5
log distance	11.9
blocked fraction	9.6
length over buildings	6.3
horizontal distance	2.4
built fraction, 50 m	1.8
receiver building height	1.7
site height	1.5

Table 14: Cell edges and dimensioning (mean over held-out cities). Disp.: mean distance between true and predicted cell edges; length: relative error of the number of cell-edge pixels. Plan: sites chosen greedily on the predicted map until it covers 90% of the pixels at 140 dB, and the true coverage of those sites; truth: sites the same procedure needs on the true map.

	Cell edges		Plan for 90%	
Predictor	disp. (m)	length (%)	sites	true cov. (%)
Tile labels (oracle)	76	−92	1.1	47.8
Source median	81	−92	1.0	54.8
3GPP UMa NLoS	81	−92	5.6	79.4
Kernel (NW)	81	−92	1.0	50.8
GP-ARD	81	−92	1.0	47.6
Ridge	81	−92	1.0	48.6
GBDT	81	−92	1.0	47.9
GBDT + 10 tiles	81	−92	1.0	48.4
Link transfer	7	−8	3.6	80.4
Link, regressor only	7	+5	1.0	54.9
Link + 25 tiles	8	−8	3.6	80.4
Truth	0	0	6.1	90+

site plan on this corpus.

5.6. *Mixed-path composition*

A tile model need not describe a link by the parameters of the receiver tile alone: the framework of [11] composes the loss from every tile the path crosses, Eq. (39), and larger tiles than the 100 m of this corpus. This subsection tests whether either choice closes the gap to link transfer, with exact and with transferred labels.

Composing the tile parameters along each path lifts the nearest-site restriction, and with exact labels it helps (Table 17, Fig. 13a): the path-averaged composition, fitted to the links of each city, has a path-loss error of 11.1 dB against 11.6 dB for the receiver-tile labels, selects the correct serving site for 54.1% of the pixels against 50.1%, places the cell edges 26 m from the true ones and loses 5.1% of the pixels when planning five sites. Most of the gain does not survive transfer. With parameters transferred by GBDT the path-averaged composition has a path-loss error of 13.3 dB (receiver-tile transfer: 12.8 dB; better in 2 of 11 cities) and a serving-site accuracy of 45.1%, above the nearest-site rule in 1 of 11 cities: every link now accumulates the transfer errors of all the tiles it crosses, and these errors decide which site appears strongest. Its planning regret is 17.3% against 18.4%, still far above the 0.7% of link transfer. The piecewise composition gains little over the receiver tile even with exact labels (11.6 dB, 49.1%), and with transferred exponents it is the worse of the two (GBDT 16.2 dB, 34.3%): an error in the exponent of one tile shifts the loss of every link that passes through it.

Larger tiles, as in [11], do not change the picture (Table 18, Fig. 13b): with tiles of 200 m to 1000 m the path-averaged and piecewise compositions have errors of 11.8 to 13.0 dB and 12.2 to 13.1 dB with exact labels, and 13.3 to 13.9 dB and 14.8 to 15.5 dB with GBDT. Over all compositions and tile sizes, the best tile model with exact labels reaches 11.1 dB and 54.1% of the serving sites, and link transfer is better than it in both; against the path-averaged composition with exact labels, link transfer is better in 11 of 11 cities in path loss and in 11 in

Table 15: Decomposition of the gap. Receiver-tile variants of the tile model with exact labels, the source median and GBDT transfer, each without and with the hurdle of the link model (+hurdle) and with a dual-slope line-of-sight/obstructed label; non-learned per-link baselines with the deterministic hurdle "loss ≤ cut-off"; and link transfer. Means over the held-out cities; cens.: censored RMSE; reach: correctly predicted reached/unreached pixels; cov.: coverage agreement at 130 dB; regret: $K = 5$.

	Path loss (dB)						
Predictor	RMSE	cens.	reach (%)	serv. (%)	cov. (%)	SINR (dB)	regret (%)
Receiver tile, exact labels							
Tile labels (oracle)	11.64	13.94	95.2	50.1	87.2	7.89	9.2
Tile labels + hurdle	11.64	10.53	95.8	50.4	89.0	7.24	0.8
Tile labels, dual slope	10.02	12.71	95.2	61.5	87.2	7.38	9.5
Tile labels, dual slope + hurdle	10.02	9.46	95.8	61.8	88.9	6.70	0.7
Receiver tile, source median							
Source median	14.51	15.89	95.2	50.1	83.9	7.68	29.8
Source median + hurdle	14.51	11.92	95.8	50.4	87.1	7.65	0.7
Source median, dual slope	11.47	13.86	95.2	62.6	84.9	7.08	33.1
Source median, dual slope + hurdle	11.47	10.09	95.8	62.8	88.1	6.94	0.7
Receiver tile, GBDT transfer							
GBDT	12.75	14.52	95.2	50.1	85.0	7.93	18.4
GBDT + hurdle	12.75	11.19	95.8	50.4	87.6	7.48	0.9
GBDT, dual slope	10.97	13.12	95.2	62.0	85.3	7.19	14.9
GBDT, dual slope + hurdle	10.97	9.98	95.8	62.2	87.6	6.65	0.9
Per-link physics, no training							
Free space + knife edge	14.66	11.16	92.8	53.8	73.1	8.98	1.5
Free space + Deygout	21.66	11.33	91.3	55.9	70.3	9.02	0.1
3GPP UMa, geometric LoS	17.88	12.21	95.2	62.3	81.5	7.74	3.1
Site-aware link model							
Link transfer	9.32	9.21	95.8	68.4	90.0	5.74	0.7

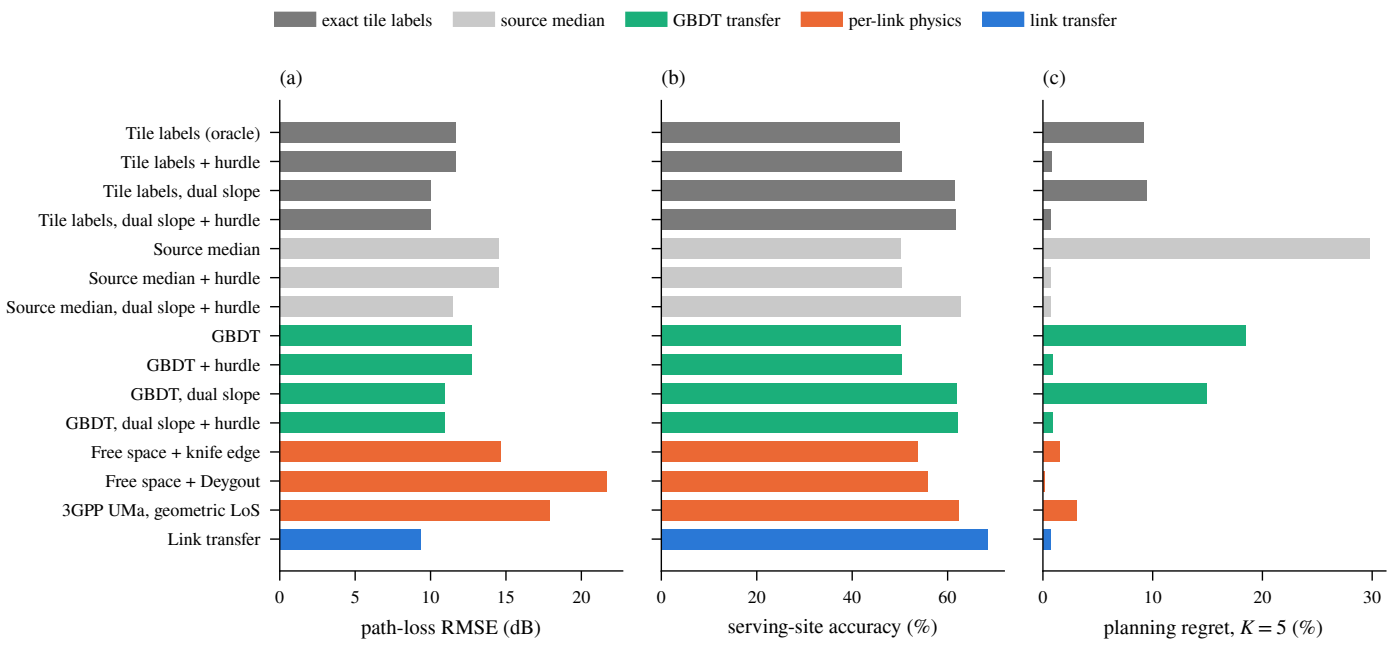


Figure 12: Decomposition of the gap (means over the held-out cities). The receiver-tile model with exact labels, the source median and GBDT transfer, each without and with the hurdle (+h) and with the dual-slope label (+LoS); the per-link physics baselines with the deterministic hurdle; and link transfer. (a) Path-loss RMSE. (b) Serving-site accuracy. (c) Planning regret for $K = 5$.

serving site, and at least as good in planning regret in 11. The limitation lies in describing a link by tile parameters, however they are combined and whatever the tile size.

For completeness, the summed reading of the composition is also reported. Algorithm 3 of [11] accumulates, for every tile a path crosses, a term $A_t + 10\gamma_t \log_{10} L_t$ with L_t the length of the path inside the tile in metres and A_t the intercept of the tile at 1 m. Read literally as a propagation law, the sum adds one intercept per crossed tile, so the loss grows with the number of tiles crossed rather than with the logarithm of the distance. With parameters fitted to the links of each city by the same joint least squares as the other two readings, its fit residual is 30.0 to 36.2 dB and the transferred parameters give 73.2 to 92.7 dB (Table 17; 40.0 to 42.7 dB with exact labels at the tile sizes

Table 16: Dual-slope labels. Share of line-of-sight links, tiles with at least 30 links of each kind (the others fall back to the all-link pair), and median parameters per kind.

	LoS links	Tiles with a pair		med. γ		med. P_{L_0} (dB)	
City	(%)	LoS	NLoS	LoS	NLoS	LoS	NLoS
Amsterdam	11.1	440/948	948/948	2.46	2.62	94.9	103.7
Barcelona	12.8	373/942	941/942	2.20	2.26	102.6	108.1
Berlin	10.9	672/957	957/957	2.60	2.61	82.0	101.8
Chicago	24.6	772/961	961/961	2.50	2.52	81.0	104.2
Hamburg	19.4	726/961	961/961	2.52	2.76	81.2	99.1
London	22.4	475/940	940/940	2.27	2.20	88.5	106.4
Munich	11.4	473/950	950/950	2.58	2.77	88.8	103.0
New York	14.2	581/960	960/960	2.57	2.41	83.0	106.6
Paris	13.2	326/910	910/910	2.12	2.26	104.3	108.5
Prague	9.8	575/960	960/960	2.62	2.76	82.7	102.3
Vienna	6.1	253/939	939/939	2.20	2.25	107.7	110.3

Table 17: Receiver-tile and mixed-path compositions of the tile model, and link transfer, on the held-out cities (mean over cities). Tile labels: fitted to the links of the held-out city itself; median, kernel and GBDT: transferred from the training cities. PL: path-loss RMSE; serv.: serving-site accuracy; SINR: mean absolute error; edge: cell-edge displacement; regret: $K = 5$.

Predictor	PL (dB)	serv. (%)	SINR (dB)	edge (m)	regret (%)
Receiver tile					
Tile labels (oracle)	11.64	50.1	7.89	76	9.2
GBDT	12.75	50.1	7.93	81	18.4
Mixed path, path-averaged					
Tile labels (oracle)	11.14	54.1	7.45	26	5.1
Source median	14.25	50.1	8.15	81	29.1
Kernel (NW)	13.41	44.5	8.08	48	18.9
GBDT	13.32	45.1	8.14	47	17.3
Mixed path, piecewise (multi-slope)					
Tile labels (oracle)	11.63	49.1	7.55	29	13.8
Source median	14.12	50.3	8.01	80	29.3
Kernel (NW)	16.27	33.2	8.18	64	26.8
GBDT	16.25	34.3	8.21	64	22.3
Mixed path, summed (literal reading)					
Tile labels (oracle)	33.20	21.3	12.11	14	7.1
Source median	92.68	48.7	7.78	53	29.3
Kernel (NW)	73.20	48.8	8.76	45	29.2
GBDT	74.09	41.4	10.44	37	20.6
Site-aware link model					
Link transfer	9.32	68.4	5.74	7	0.7

of Table 18). The published framework was evaluated against measurements by the normalised error of the received power and may normalise the accumulated terms in a way its text does not state; the piecewise reading, which the same accumulation defines once the intercept is counted once, is treated here as its faithful form, and no conclusion about the framework is drawn from the summed reading.

5.7. Survey design and certified coverage

Two instruments of Section 4 connect the link model to the operation of a digital twin: the calibration of its per-site offsets from a small survey of the target city, Eq. (26), and the city-level coverage certificate of Proposition 4. Both are evaluated here on the held-out cities; the survey is revisited on the external scenes in Section 5.8, where it matters most.

Calibrating the per-site offsets of the link model from surveyed tiles changes it little on the corpus (Table 19, Fig. 14, scored on the unsurveyed pixels). With 25 random tiles and the empirical-Bayes κ of Eq. (23) the path-loss error goes from 9.32 dB to 9.35 dB (+0.03 dB, [+0.02, +0.04], better in 1 of 11 cities) and the serving-site accuracy from 68.4% to 66.4%, with the regret unchanged at 0.7%; the estimated κ has a median of 37 links and reaches its ceiling, which switches the per-site offsets off, in 0% of the draws. The fixed $\kappa = 50$ links gives 9.34 dB (+0.00 dB against the empirical-Bayes value, [−0.01, +0.00]). The variance design of Proposition 3 gives 9.56 dB (+0.21 dB against random tiles, [+0.16, +0.27]) and the stratified design 9.41 dB (+0.07 dB, [+0.03, +0.12]); the best of the four variants at $k = 25$ is random tiles with the fixed κ. The zero-shot link model has little systematic per-site error on this corpus for a survey of this size to remove; a design that concentrates the survey on the sites with the fewest links selects unrepresentative tiles and makes the model worse.

Table 20 and Fig. 15 apply Proposition 4. For every predictor the bound $\hat{q}_t$ holds in 10 of 11 cities. This count is fixed by construction (with distinct margins, the bound of rank j fails exactly in the j cities with the largest q_c) and is a consistency check, not evidence; the guarantee concerns a new city drawn exchangeably with the eleven. At $\alpha = 0.1$ and 140 dB, the certificate of link transfer needs a margin of 13.9 dB and certifies 80.6% of the pixels, with 1.31% falsely claimed; the true covered share is 91.8%. Tile transfer needs larger margins and certifies less: GBDT 48.4% (margin 24.1 dB), the source median 20.1% and even the true tile labels 54.3%. The naive claim without a margin covers almost every pixel but is wrong for 4.89% (link) to 8.20% (GBDT) of them, with no stated risk (Fig. 15b). Using the second-largest margin of the other cities ($j = 2$) lowers the confidence to 9/11 and raises the certified share of link transfer to 82.6% (GBDT 49.8%); the bound then holds in 9 of 11 cities. The falsely claimed shares stay far below α because the margin must cover the worst city; the value of the certificate is the guarantee itself, and with eleven cities its resolution is coarse: a 95% level would need twenty cities.

5.8. Robustness

The main results rest on one tile size, one cut-off, one network density, one siting rule, one material model and one band. This subsection varies each of them in turn, following Section 4.7, and closes with the external scenes, which were modelled independently of the corpus.

Table 21 repeats the tile pipeline for $G_s \in \{50, 100, 200\}$ m; $G_s = 100$ m reproduces the main results exactly. Larger tiles give more precise labels (cluster-robust standard error 0.540, 0.456 and 0.369) and a smaller exponent error (GBDT 0.872, 0.764 and 0.612), but a coarser map: the path-loss error of the oracle goes from 11.2 dB to 12.1 dB, and that of GBDT tile transfer from 12.7 dB to 13.0 dB. The serving-site accuracy of the tile model is 50.1% at every size, as the nearest-site rule implies. No tile size approaches link transfer. The labels also depend on the dynamic range of the data (Table 22). Between 36.0% and 69.9% of the (pixel, site) pairs of the labelled tiles lie beyond the cut-off; lowering the cut-off to 140 dB changes the median tile exponent by −0.583 and to 130 dB by −1.119, and a Tobit fit that also uses the censored pairs gives slopes larger than the least-squares slopes by a median of 2.11 to 4.49

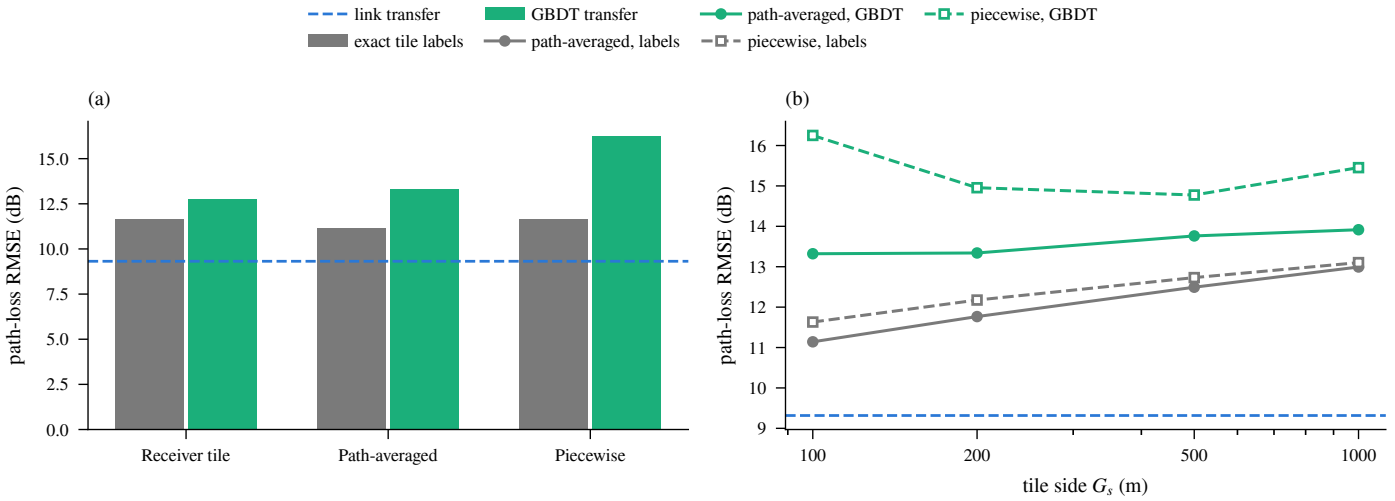


Figure 13: Compositions of the tile model against link transfer (path-loss RMSE, mean over held-out cities). (a) Receiver-tile, path-averaged and piecewise compositions at G_s = 100 m with exact labels and with GBDT transfer. (b) Path-averaged and piecewise compositions against the tile side, with exact labels and with GBDT transfer. The dashed line is link transfer.

Table 18: Mixed-path compositions of the tile model for tile sides G_s of 100 m (the corpus) and 200 m, 500 m and 1000 m (the range of [11]), on the pixels of the main corpus (mean over held-out cities). Tiles: tiles per city crossed by at least 30 links; labels: parameters fitted to the held-out city itself; GBDT: transferred from the training cities. PL: path-loss RMSE (dB); serv.: serving-site accuracy (%). Link transfer: 9.3 dB, 68.4%.

		Summed (literal)				Piecewise				Path-averaged			
		labels		GBDT		labels		GBDT		labels		GBDT	
G_s (m)	tiles	PL	serv.	PL	serv.	PL	serv.	PL	serv.	PL	serv.	PL	serv.
100	951	33.2	21.3	74.1	41.4	11.6	49.1	16.2	34.3	11.1	54.1	13.3	45.1
200	241	40.0	29.7	61.5	41.4	12.2	49.3	15.0	38.3	11.8	51.5	13.3	46.3
500	49	42.6	33.7	49.7	35.1	12.7	47.4	14.8	36.9	12.5	47.7	13.8	46.7
1000	14	42.7	37.7	47.3	44.5	13.1	47.0	15.5	46.6	13.0	48.5	13.9	48.5

Table 19: Per-site calibration of the link model from k surveyed tiles, scored on the unsurveyed pixels: random tiles (mean of ten draws) with the empirical-Bayes κ and with the fixed κ = 50, the variance design of Proposition 3 and the stratified design. PL: path-loss RMSE (dB); serv.: serving-site accuracy (%); regret: K = 5 (%). Means over the held-out cities. Zero-shot link model: 9.32 dB, 68.4%, 0.7%.

	Random, EB κ			Random, fixed κ			Variance design			Stratified design		
k	PL	serv.	regret	PL	serv.	regret	PL	serv.	regret	PL	serv.	regret
10	9.49	64.9	0.7	9.45	65.2	0.7	9.56	64.9	0.7	9.50	64.2	0.9
25	9.35	66.4	0.7	9.34	66.4	0.7	9.56	64.7	0.8	9.41	65.3	0.7
50	9.30	67.0	0.7	9.30	67.0	0.7	9.61	64.5	0.8	9.36	66.1	0.7

across the cities (3,300 tiles). The tile exponent thus depends on the dynamic range of the measurements, not only on the environment, which is a further reason to model links, and to model the cut-off explicitly as the hurdle of the link model does. On the network of the first 10 sites (Table 23), the ranking is unchanged: the oracle has a path-loss error of 11.7 dB, GBDT tile transfer 12.9 dB and link transfer 9.2 dB, with serving-site accuracies of 52.8%, 52.9% and 67.8%.

Table 24 and Fig. 16 apply the models of the main corpus, unchanged, to the 3 replicates (random rooftops, mid-rise rooftops, random materials) of 4 cities (Amsterdam, London, New York, Paris). With sites on random rooftops (median site height 26 m against 55 m on the main corpus) the link models reach a path-loss error of 10.8 dB, against 12.0 dB for the tile model with the exact labels of the new network, with a planning regret of 2.0% against 13.5%. With sites on mid-rise rooftops (median 26 m, the 3GPP urban-macro range) they reach 11.1 dB against 12.1 dB for the exact labels, with a regret of 0.9% against 12.0%; the link model trained on tall sites is therefore not tied to them. With random building materials the truth shifts by +5.3 dB relative to the concrete-trained link models, whose error is 11.1 dB against 10.6 dB for the exact labels of that network (the link model is better than them in 1 of 4 cities) and 13.6 dB for GBDT tile transfer; the shift is the systematic error that the survey of Section 5.7 is for. The tile labels of the same tiles fitted under the main corpus give 15.9 dB,

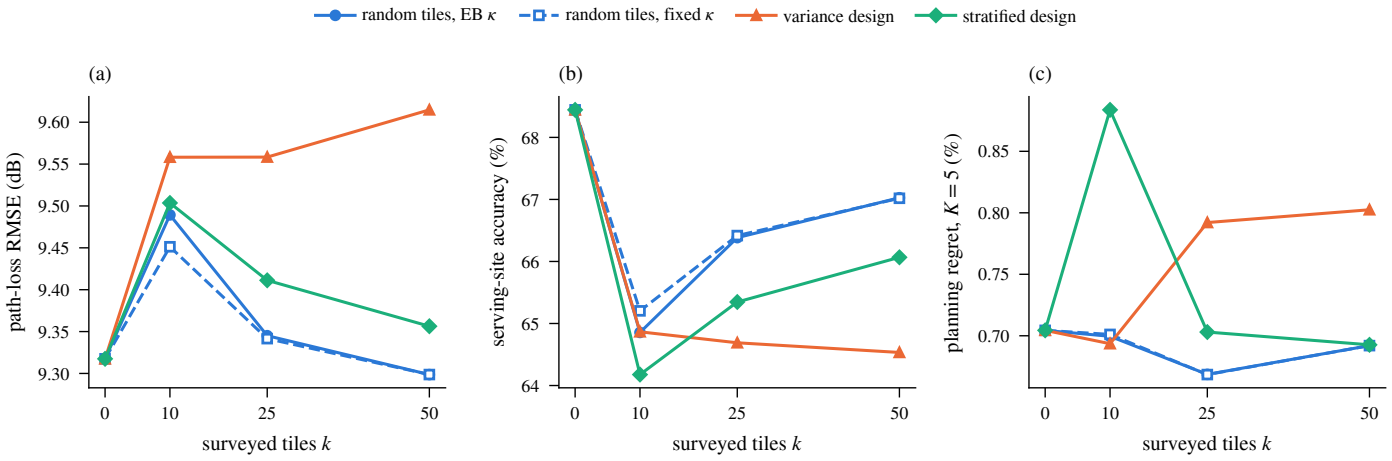


Figure 14: Per-site calibration of the link model against the number of surveyed tiles k (mean over held-out cities, unsurveyed pixels): random tiles with the empirical-Bayes and with the fixed κ, the variance design and the stratified design. (a) Path-loss RMSE. (b) Serving-site accuracy. (c) Planning regret for $K = 5$.

Table 20: Certified coverage (Proposition 4), mean over held-out cities. $\hat{q}_t$: margin from the other cities; valid: cities in which $q_t \le \hat{q}_t$ (of 11); $r_p \le \hat{q}_t$: share of pixels within the margin; certified claim: share claimed covered and share claimed but not covered; naive claim: without a margin; covered: true covered share.

				Certified claim		Naive claim		
Predictor	$\hat{q}_t$ (dB)	valid	$r_p \le \hat{q}_t$ (%)	claimed (%)	false (%)	claimed (%)	false (%)	covered (%)
$\alpha = 0.1$, $L = 140$ dB, *bound holding with probability* $\ge 10/11$								
Tile labels (oracle)	22.7	10	96.4	54.3	0.56	99.8	8.05	91.8
Source median	30.3	10	96.6	20.1	0.52	100.0	8.20	91.8
3GPP UMa NLoS	21.6	10	96.4	29.0	0.87	95.3	7.72	91.8
Kernel (NW)	25.7	10	96.4	42.2	0.45	100.0	8.20	91.8
Ridge	23.9	10	96.1	49.5	0.64	100.0	8.20	91.8
GBDT	24.1	10	95.9	48.4	0.59	100.0	8.20	91.8
Link transfer	13.9	10	94.9	80.6	1.31	95.3	4.89	91.8
Link, regressor only	16.4	10	95.4	75.1	0.74	100.0	8.20	91.8
$\alpha = 0.1$, $L = 140$ dB, *bound holding with probability* $\ge 9/11$								
Tile labels (oracle)	20.7	9	94.8	61.2	1.13	99.8	8.05	91.8
Source median	29.0	9	95.8	24.5	0.72	100.0	8.20	91.8
3GPP UMa NLoS	20.2	9	95.6	33.4	1.13	95.3	7.72	91.8
Kernel (NW)	25.2	9	95.8	44.1	0.58	100.0	8.20	91.8
Ridge	22.5	9	94.8	54.5	0.97	100.0	8.20	91.8
GBDT	23.8	9	95.3	49.8	0.74	100.0	8.20	91.8
Link transfer	13.0	9	94.4	82.6	1.67	95.3	4.89	91.8
Link, regressor only	15.9	9	94.8	76.2	0.84	100.0	8.20	91.8
$\alpha = 0.05$, $L = 140$ dB, *bound holding with probability* $\ge 10/11$								
Tile labels (oracle)	26.3	10	98.2	41.9	0.21	99.8	8.05	91.8
Source median	34.0	10	98.4	10.7	0.18	100.0	8.20	91.8
3GPP UMa NLoS	26.4	10	98.3	17.0	0.36	95.3	7.72	91.8
Kernel (NW)	29.4	10	98.4	28.7	0.14	100.0	8.20	91.8
Ridge	27.4	10	98.1	36.4	0.26	100.0	8.20	91.8
GBDT	27.6	10	98.1	35.9	0.22	100.0	8.20	91.8
Link transfer	18.7	10	97.5	69.8	0.40	95.3	4.89	91.8
Link, regressor only	19.3	10	97.6	68.4	0.34	100.0	8.20	91.8

16.1 dB and 12.1 dB on the three replicates: the labels of a tile do not carry over to other sites or materials, as the dependence of the exponent on the serving sites (Section 5.2) implies.

At 7.5 GHz the corpus has 7,038,989 links below the cut-off (10,274 labelled tiles). The tile exponents are lower than at 3.5 GHz (median 2.18 against 2.79, Fig. 17a) and the intercepts higher by a median of 7.9 dB, against 6.6 dB for free space; the per-tile exponents of the two bands correlate with $r = 0.74$–0.92 across cities. The lower exponents agree with the stronger censoring of the higher band. The ranking of the predictors is unchanged (Table 25, Fig. 17b): the tile labels give 10.9 dB, GBDT tile transfer 11.9 dB and link transfer trained at 7.5 GHz

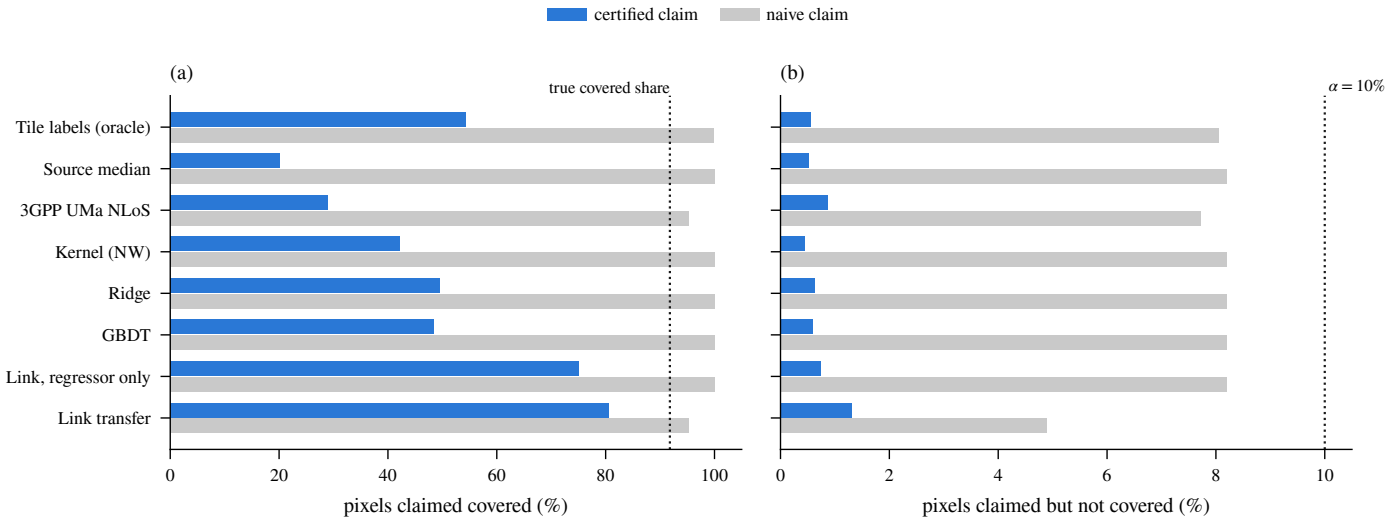


Figure 15: Certified coverage at $\alpha = 0.1$, $L = 140$ dB and rank $j = 1$ (mean over held-out cities). (a) Share of pixels claimed covered under the certificate and under the naive claim without a margin; the dotted line is the true covered share. (b) Share of pixels claimed but not covered; the dotted line is the risk level α.

Table 21: Tile size. Tiles per city, median shrinkage weight (mean of the city medians) and standard error of the exponent (iid and cluster-robust, RMS over all tiles); zero-shot exponent RMSE of the source median and GBDT; network path-loss RMSE of the tile labels (oracle) and of GBDT tile transfer; serving-site accuracy of the tile model. Means over the held-out cities.

			se(γ)		γ RMSE		PL RMSE (dB)		
G_s (m)	tiles/city	med. w_t	iid	cluster	median	GBDT	oracle	GBDT	serv. (%)
50	3470	0.38	0.234	0.540	0.985	0.872	11.19	12.66	50.1
100	948	0.37	0.141	0.456	0.871	0.764	11.64	12.75	50.1
200	244	0.38	0.088	0.369	0.713	0.612	12.09	12.98	50.1

Table 22: Path-loss cut-off. Share of (pixel, site) pairs of the labelled tiles beyond the 150 dB cut-off and of links above 140 dB; median change of the tile exponent when the cut-off is lowered to 140 and 130 dB; for random tiles, median and 90th percentile of the Tobit minus the least-squares slope.

	censored	Lower cut-off			Tobit vs least squares		
City	pairs (%)	links >140 (%)	$\Delta\gamma_{140}$	$\Delta\gamma_{130}$	tiles	med. $\Delta\gamma$	q90
Amsterdam	59.0	18.4	−0.572	−1.141	300	+3.746	+6.883
Barcelona	65.7	18.7	−0.486	−0.893	300	+4.278	+7.432
Berlin	43.9	17.3	−0.646	−1.261	300	+3.256	+6.908
Chicago	40.6	17.9	−0.640	−1.226	300	+3.527	+6.698
Hamburg	36.0	15.0	−0.504	−1.066	300	+2.113	+3.886
London	48.2	14.6	−0.470	−0.855	300	+2.924	+5.013
Munich	55.9	20.1	−0.583	−1.184	300	+3.131	+5.333
New York	54.0	23.8	−0.621	−1.119	300	+4.492	+7.342
Paris	69.3	18.3	−0.598	−1.006	300	+4.235	+8.410
Prague	56.4	21.5	−0.729	−1.366	300	+3.500	+5.315
Vienna	69.9	22.9	−0.527	−0.980	300	+4.106	+7.879

8.5 dB (better than GBDT in 11 of 11 cities). The link models trained at 3.5 GHz transfer across bands with the free-space offset alone: 9.0 dB, and 9.1 dB with the knife-edge feature evaluated at the new wavelength, both below the tile model with the exact labels of the new band, with a serving-site accuracy of 68.8% and a planning regret of 1.2%.

On the 3 independently modelled scenes (Table 26, Fig. 18; 202,782 links), whose geometry and materials differ from the training corpus, zero-shot link transfer remains better than transferred tile parameters in path loss (13.1 dB against 14.8 dB for GBDT, in 3 of 3 scenes), serving site (68.3% against 55.4%), SINR (5.0 dB against 5.3 dB) and cell edges (9 m against 45 m), but its path-loss error is above that of the tile model with the exact labels of these scenes (12.24 dB) and it has a bias of +2.9 to +4.7 dB (positive in every scene), consistent with the different materials of these scenes against the single concrete model of the training corpus. This is the systematic shift that the per-site calibration of Section 4.4 is for:

Table 23: Network size: the same predictors on the 20-site and on the 10-site network (mean over held-out cities). PL: path-loss RMSE; serv.: serving-site accuracy; SINR: mean absolute error; regret: $K = 5$.

	20 sites				10 sites			
Predictor	PL (dB)	serv. (%)	SINR (dB)	regret (%)	PL (dB)	serv. (%)	SINR (dB)	regret (%)
Tile labels (oracle)	11.64	50.1	7.89	9.2	11.67	52.8	8.46	5.5
Source median	14.51	50.1	7.68	29.8	14.90	52.9	8.40	26.9
3GPP UMa NLoS	22.75	50.1	7.56	9.0	23.87	52.9	8.63	4.1
Kernel (NW)	12.91	50.1	7.92	24.3	13.08	52.9	8.57	19.8
GP-ARD	12.77	50.1	7.92	15.2	12.92	52.9	8.56	13.8
Ridge	12.76	50.1	7.96	19.9	12.92	52.9	8.58	15.8
GBDT	12.75	50.1	7.93	18.4	12.90	52.9	8.56	13.7
Link transfer	9.32	68.4	5.74	0.7	9.21	67.8	6.43	0.7

Table 24: Siting and materials replicates: models of the main corpus applied to 4 cities (Amsterdam, London, New York, Paris) re-traced with sites on random rooftops, with sites on mid-rise rooftops (12–45 m) and with random ITU materials per building. Left: the same predictors on the main corpus of these cities. Tile labels of this network: fitted on the replicate's links; main corpus: the labels of the same tiles in the main corpus. PL: path-loss RMSE (dB); serv.: serving-site accuracy (%); regret: $K = 5$ (%). Means over cities (and seeds).

	Main corpus			Random rooftops			Mid-rise rooftops			Random materials		
Predictor	PL	serv.	regret	PL	serv.	regret	PL	serv.	regret	PL	serv.	regret
Tile labels of this network	11.71	50.1	8.2	12.04	54.3	13.5	12.09	58.5	12.0	10.64	54.7	11.9
Tile labels, main corpus	–	–	–	15.87	54.8	14.8	16.15	58.9	16.4	12.06	54.6	5.7
Source median	14.78	50.1	32.7	15.21	54.9	34.6	15.36	59.0	42.5	15.12	54.7	30.3
GBDT	12.97	50.1	17.0	14.81	54.9	27.4	14.93	59.0	25.8	13.60	54.7	16.4
Link transfer	9.33	66.4	0.9	10.82	64.8	2.0	11.05	65.6	0.9	11.07	66.8	0.3

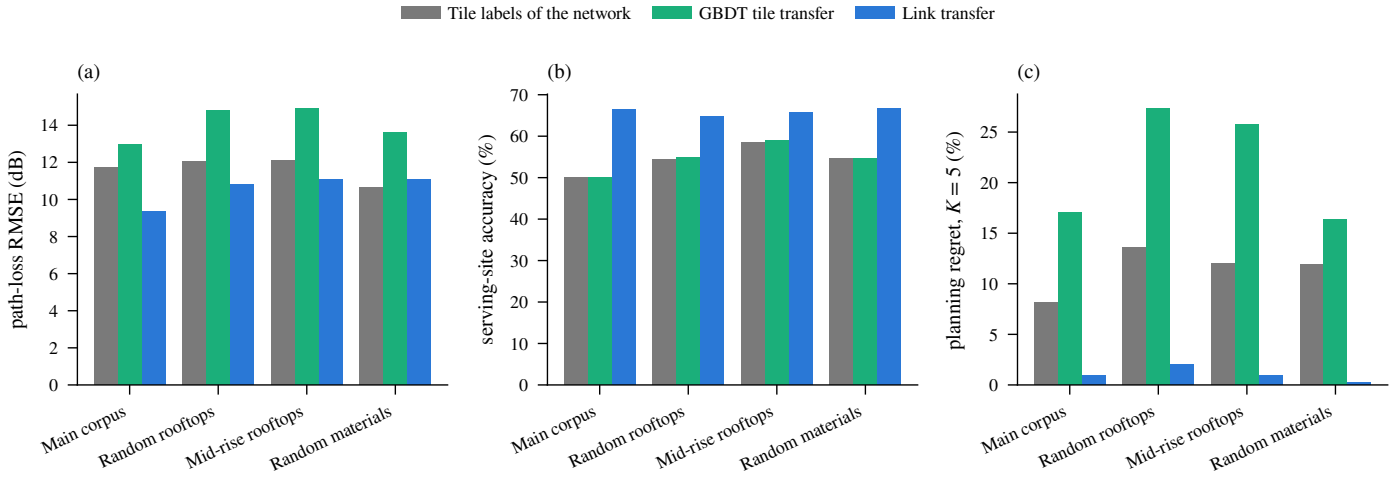


Figure 16: The main corpus and the three replicates of the 4 re-traced cities (means over cities): the tile model with the exact labels of each network, GBDT tile transfer and link transfer, the latter two trained on the main corpus only. (a) Path-loss RMSE. (b) Serving-site accuracy. (c) Planning regret for $K = 5$.

with 25 randomly surveyed tiles the bias becomes −0.1 dB on average and the path-loss error 12.35 dB on the unsurveyed pixels, against 12.24 dB for the exact tile labels (lower in 1 of 3 scenes), while the serving-site accuracy becomes 63.5% against 55.4% for the tile labels. The variance design of 25 tiles reaches 14.02 dB and the stratified design 12.83 dB. With 8 sites on these smaller scenes, five sites cover almost every pixel, and the zero-shot predictors have identical planning regrets within each scene (1.6–2.2% across the scenes).

5.9. Statistical summary

The statements of the preceding subsections rest on 130 city-level comparisons computed as described in Section 4.7. Table 27 lists the key comparisons behind the main claims; the complete list is part of the released result files.

Link transfer improves on GBDT tile transfer in path loss in 11, in serving site in 11, in SINR in 11 and in planning regret in 11 of 11 cities, and on the tile model with the true labels in path loss in 11, in serving site in 11 and in planning regret in 11 of 11 cities; these comparisons are significant after Holm's

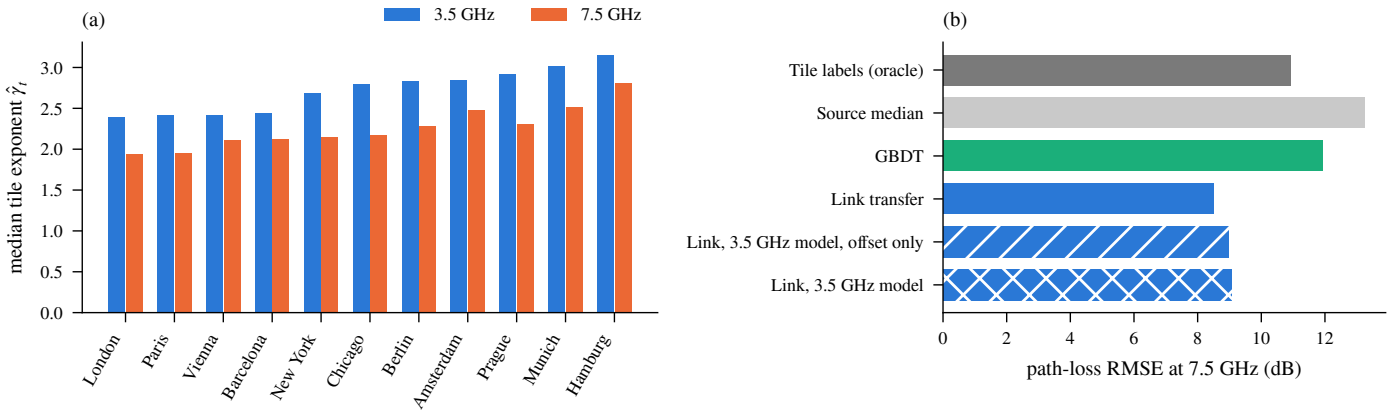


Figure 17: Second band. (a) Median tile exponent per city at 3.5 GHz and 7.5 GHz. (b) Path-loss RMSE at 7.5 GHz (mean over held-out cities) of the tile labels, the source median, GBDT tile transfer and link transfer trained at 7.5 GHz, and of the 3.5 GHz link models applied across bands with the free-space offset only and with the knife-edge feature at the new wavelength (hatched).

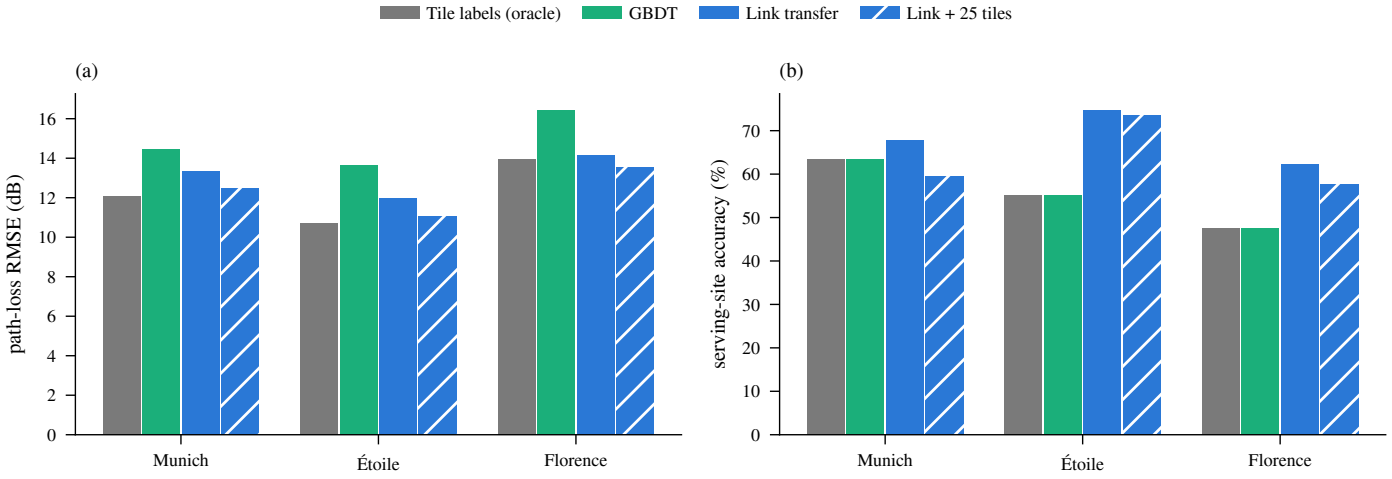


Figure 18: External scenes: the tile model with the exact labels of each scene, GBDT tile transfer, zero-shot link transfer and link transfer calibrated from 25 random surveyed tiles (hatched; unsurveyed pixels). (a) Path-loss RMSE. (b) Serving-site accuracy.

Table 25: Second band (7.5 GHz), mean over held-out cities. Above the rule: models trained at 7.5 GHz; below: the link models trained at 3.5 GHz with the free-space offset, with the knife-edge feature at the new wavelength or unchanged.

Predictor	PL (dB)	serv. (%)	SINR (dB)	regret (%)
Tile labels (oracle)	10.92	49.6	8.76	7.1
Source median	13.25	49.8	8.77	33.0
3GPP UMa NLoS	24.88	49.8	8.93	7.9
GBDT	11.91	49.8	8.73	15.6
Link transfer	8.51	65.6	6.45	0.7
Link, 3.5 GHz model	9.07	68.8	6.73	1.2
Link, 3.5 GHz model, offset only	8.98	68.8	6.70	1.2

correction. The decomposition comparisons of Section 5.5 and the second-band comparison of Section 5.8 behave alike. At the parameter level the evidence is weaker: of the 12 comparisons of a transfer operator with the source median, 0 improvements survive Holm's correction at eleven cities, whereas the gain of GP-ARD over kernel transfer is significant, and the few-shot gain of ten surveyed tiles is not significant. With eleven cities an exact Wilcoxon test can reject only when nearly every city agrees, so the city counts of Table 27 are the more informative column for the parameter-level comparisons.

6. Discussion

The results of Section 5 bear on three questions: why the tile exponent transfers poorly across cities, what the gap between tile models and link transfer consists of, and what follows for the propagation layer of a digital twin network. This section takes them in turn, returns to the kernel operator, draws the practical conclusions for planning and states the limitations of the study.

The central premise of descriptor-based tile transfer is that a tile has a path-loss exponent determined by its environment, which can be read from descriptors and carried to similar tiles elsewhere. On this corpus the premise is weak, for three measured reasons. Two estimates of the exponent of the same tile from disjoint halves of the sites differ by 1.37 in standard deviation, so the exponent depends on which sites serve the tile;

Table 26: External scenes of Sionna RT (no retraining). PL: path-loss RMSE; bias: mean path-loss error, predicted minus true (dB); serv.: serving-site accuracy; SINR: mean absolute error; edge: cell-edge displacement; regret: $K = 5$. Tile labels: fitted on the scene itself; median, ridge and GBDT: fitted on the eleven cities; link: mean of the eleven leave-one-city-out link models, zero-shot and calibrated from 25 surveyed tiles (random: mean of ten draws; the variance design; the stratified design), scored on the unsurveyed pixels.

Predictor	PL (dB)	bias	serv. (%)	SINR (dB)	edge (m)	regret (%)
Munich (104,696 links, 8 sites, 195 tiles)						
Tile labels (oracle)	12.09	+0.0	63.4	5.55	48	1.6
Source median	15.33	+7.1	63.4	4.56	48	1.6
Ridge	14.64	+3.5	63.4	4.55	48	1.6
GBDT	14.43	+2.8	63.4	4.57	48	1.6
Link transfer	13.33	+4.2	67.8	4.47	10	1.6
Link + 25 tiles	12.47	+0.5	59.3	5.26	12	1.6
Link + 25 tiles, variance design	14.69	+7.4	49.1	5.56	12	1.7
Link + 25 tiles, stratified design	12.71	−1.9	62.3	5.00	11	1.4
Étoile (42,235 links, 8 sites, 63 tiles)						
Tile labels (oracle)	10.69	+0.0	55.1	4.44	38	2.0
Source median	15.12	+8.5	55.1	4.67	38	2.0
Ridge	13.87	+2.4	55.1	4.85	38	2.0
GBDT	13.63	+2.0	55.1	4.83	38	2.0
Link transfer	11.97	+4.7	74.8	4.38	6	2.0
Link + 25 tiles	11.04	−0.8	73.6	4.41	7	1.9
Link + 25 tiles, variance design	12.88	−5.5	76.7	4.87	7	1.9
Link + 25 tiles, stratified design	12.26	−4.4	74.3	4.68	7	2.4
Florence (55,851 links, 8 sites, 127 tiles)						
Tile labels (oracle)	13.93	+0.0	47.6	7.37	44	2.2
Source median	15.51	+1.1	47.6	6.52	48	2.2
Ridge	16.70	+0.2	47.6	6.69	48	2.2
GBDT	16.44	−1.0	47.6	6.61	48	2.2
Link transfer	14.14	+2.9	62.3	6.02	10	2.2
Link + 25 tiles	13.55	+0.0	57.5	6.89	14	2.4
Link + 25 tiles, variance design	14.49	+6.2	54.6	7.07	15	2.5
Link + 25 tiles, stratified design	13.53	−1.7	58.8	6.69	10	1.0
Mean over scenes						
Tile labels (oracle)	12.24	+0.0	55.4	5.79	43	1.9
Source median	15.32	+5.6	55.4	5.25	45	1.9
Ridge	15.07	+2.0	55.4	5.37	45	1.9
GBDT	14.84	+1.2	55.4	5.34	45	1.9
Link transfer	13.14	+3.9	68.3	4.95	9	1.9
Link + 25 tiles	12.35	−0.1	63.5	5.52	11	2.0
Link + 25 tiles, variance design	14.02	+2.7	60.1	5.83	11	2.0
Link + 25 tiles, stratified design	12.83	−2.7	65.1	5.46	9	1.6

it depends on the dynamic range of the measurements, shifting by −0.583 when the cut-off is lowered by 10 dB; and the cross-city spread of the city medians (0.27) is small against the spread within a city. A tile is seen by each site through a different street canyon, and the exponent fitted over all of them mixes the geometry of the tile with that of the paths reaching it. This explains why learned operators improve on the source median only modestly, why few-shot correction of the exponent gains only 0.6% with ten surveyed tiles and 1.9% with 100, and why a planning tool fed with even the *true* receiver-tile labels assigns half of the pixels to the wrong site and draws cell edges 76 m away from the true ones.

Section 5.5 separates three things that a receiver-tile model lacks. The first is a notion of reach: without a cut-off model every site covers every pixel, a single site appears to cover 90% of a city, and a plan built on such a map loses 18.4% of the pixels; the hurdle of the link model, applied to the same tile map, brings the regret to 0.9%. The second is the line-of-sight state: a dual-slope label raises the serving-site accuracy from 50.1% to 62.0%. The third is the geometry of the individual link, the height of the obstruction above the line of sight, the buildings crossed and the length over them, which the tile representation cannot carry at any tile size and which accounts for the remaining distance to link transfer. The per-link physics baselines, which use the same geometry without any training, reach 14.7 dB at best; the learned link model is better than the best of them in 11 of 11 cities in path loss and in 11 in serving site, but not in the five-site plan, where the Deygout construction with its deterministic hurdle is already as good (0.1% against 0.7%). The learning adds accuracy to the diffraction physics; the reach model is what makes either plan reliable.

The quantity that transfers is therefore the effect of the geom-

Table 27: Key city-level comparisons ($n = 11$ held-out cities). Δ: mean difference first minus second (percentage points for shares: serving site, coverage agreement, regret); CI: 95% percentile bootstrap over cities; p: exact two-sided Wilcoxon signed-rank; better: cities in which the first is better; Holm: significant at 5% after Holm's correction within the family.

Comparison	Metric	Δ	95% CI	p	better	Holm
Zero-shot vs source median						
GBDT vs Source median	γ RMSE	−0.106	[−0.157, −0.052]	0.010	9/11	no
Kernel (NW) vs Source median	γ RMSE	−0.070	[−0.129, −0.001]	0.083	9/11	no
Zero-shot pairs						
GP-ARD vs Kernel (NW)	γ RMSE	−0.039	[−0.063, −0.021]	0.002	10/11	yes
Few-shot correction						
GBDT, k=10 vs GBDT, k=0	γ RMSE	−0.005	[−0.011, −0.001]	0.032	9/11	no
Network level						
GBDT vs Source median	path-loss RMSE (dB)	−1.75	[−2.27, −1.26]	<0.001	11/11	yes
Link transfer vs GBDT	path-loss RMSE (dB)	−3.44	[−3.83, −3.10]	<0.001	11/11	yes
Link transfer vs Tile labels (oracle)	path-loss RMSE (dB)	−2.33	[−2.63, −2.08]	<0.001	11/11	yes
Link transfer vs GBDT	serving site	+18.3	[+14.5, +22.6]	<0.001	11/11	yes
Link transfer vs GBDT	regret, K=5	−17.7	[−23.5, −12.6]	<0.001	11/11	yes
Link transfer vs Tile labels (oracle)	regret, K=5	−8.5	[−11.1, −5.9]	<0.001	11/11	yes
Link + 25 tiles vs Link transfer	path-loss RMSE (dB)	+0.03	[+0.02, +0.04]	0.003	1/11	yes
Link + 25 tiles, variance design vs Link + 25 tiles	path-loss RMSE (dB)	+0.21	[+0.16, +0.27]	<0.001	0/11	yes
Decomposition and physics baselines						
GBDT + hurdle vs GBDT	regret, K=5	−17.5	[−23.3, −12.4]	<0.001	11/11	yes
GBDT, dual slope vs GBDT	serving site	+11.8	[+9.2, +15.2]	<0.001	11/11	yes
Link transfer vs GBDT, dual slope + hurdle	path-loss RMSE (dB)	−1.65	[−1.81, −1.50]	<0.001	11/11	yes
Link transfer vs Free space + Deygout	path-loss RMSE (dB)	−12.34	[−15.83, −9.53]	<0.001	11/11	yes
Link transfer vs Free space + Deygout	regret, K=5	+0.6	[+0.2, +1.0]	0.055	3/11	no
Network level, second band						
Link transfer vs GBDT	path-loss RMSE (dB)	−3.40	[−3.78, −3.08]	<0.001	11/11	yes
Link, 3.5 GHz model vs Link transfer	path-loss RMSE (dB)	+0.55	[+0.43, +0.69]	<0.001	0/11	yes

etry along a link, not a per-tile summary of it. Site-aware link transfer uses the same descriptors, the same training cities and the same protocol, but conditions on the path between site and receiver; it improves path loss, serving site, coverage and planning in every held-out city and outperforms the tile model with perfect labels, however the tile parameters are composed along the path and whatever the tile size (Section 5.6). It keeps its advantage over transferred tile parameters at another frequency, with other sites, with other building materials and on independently modelled scenes. On the latter it is biased until a survey of 25 random tiles calibrates it, after which its path loss is close to that of the exact tile labels and its serving-site accuracy exceeds theirs: the material models of the training corpus limit what transfers zero-shot, and a small unbiased survey corrects the rest. Its most important features, the knife-edge loss and the height of the obstruction above the line of sight, are the quantities of classical diffraction theory [13], so the model remains interpretable in physical terms. Three design choices matter for planning. First, the model must represent the absence of a link: the hurdle, which predicts whether a site is received at all, reduces the planning regret from 4.5% to 0.7% at the same path-loss error over the ray-traced links. Second, a calibration from surveyed links must be allowed to do little: on this corpus the empirical-Bayes prior strength has a median of 37 links, close to the fixed value of 50, and both leave the zero-shot model nearly unchanged, whereas the designs that concentrate the survey make it worse; on the external scenes the same calibration removes a bias of several decibels. Third, its errors must be stated at the level of the decision: the city-level certificate turns the leave-one-city-out residuals into a coverage promise with a guaranteed risk, and the more accurate the model, the more of the city it can promise (80.6% against 48.4% for tile transfer). For a 6G network digital twin this means that the transferable representation should be attached to links (or to site–tile pairs), which is also what a planning tool evaluates.

The analysis of the kernel operator completes the picture at the parameter level. Proposition 2 identifies two limits of kernel transfer; the tuned operator is in neither, but with the largest bandwidth selected on 9 targets it approaches the source mean without reaching the flat regime. Its disadvantage against the learned regressors is the noise-transmission term of Eq. (7) combined with an isotropic distance: learning one length scale per descriptor (GP-ARD) removes most of the gap, which locates the weakness in the metric rather than in kernel averaging as such.

Four practical conclusions follow for the planning functions of a digital twin. First, a transfer model should be validated at the network level; parameter errors of similar size lead to very different planning losses (Table 11), and a uniformly biased model such as the 3GPP UMa NLoS constants has a path-loss error of 22.8 dB, far above all transfer operators, yet a planning regret of 9.0%, below that of transferred tile parameters, because a uniform bias does not change which sites cover the most pixels. Second, models trained on links below a receiver

cut-off overstate the reach of distant sites unless the cut-off is modelled; the tile models with transferred parameters predict that a single site covers 90% of a city (GBDT: a single site; source median: a single site), and a hurdle fixes most of the planning loss even for the tile model. Third, small surveys do not help a model without systematic error, but correct a shifted one, and they must be unbiased: a design that maximises the variance reduction of the offsets selects unrepresentative tiles, and the empirical-Bayes prior should decide how much to trust them. Fourth, coverage claims should come with a margin estimated across cities: without it, 8.20% of the pixels promised by tile transfer are not covered.

The study has limitations that bound the reach of these conclusions. The path loss is ray traced rather than measured, so absolute values depend on the solver, the ITU material models and the level-of-detail-1 geometry; the results are comparisons under identical conditions, and the solver noise floor of Section 5.2 bounds the agreement any model can reach near the cut-off. Ray tracing on single-material geometry makes path loss a near-deterministic function of the obstruction geometry that the link features encode; measured data carry variance from materials, vegetation, vehicles and indoor clutter that a per-link geometry does not see and that a tile statistic partly absorbs. The materials replicate and the external scenes test the first of these and show the expected bias, which a small survey removes; vegetation and clutter remain untested, and none of the replicates replaces validation against measurements with known site positions, which the public drive-test data available to the authors do not provide. Between 2% and 50% of the building heights are imputed per city. With eleven cities, city-level tests have limited power: an exact Wilcoxon test with Holm's correction over a dozen comparisons requires nearly every city to agree, which is why no improvement of a tile operator over the median survives it although the learned operators improve on it in 8 to 9 of 11 cities, and why the certificate cannot state a 95% level; a corpus of twenty or more cities, which the released pipeline generates at 77 s per site, would remove both limits. The corpus has flat terrain, 20 rooftop sites per city with isotropic antennas, full load and a fixed link budget. Tile labels come from the same ray tracer as the network truth, so the oracle is the reference for tile models with exact labels on this corpus, not a statement about measured data.

7. Conclusions and future work

This article has evaluated the cross-city transfer of propagation models for network planning in 6G digital twin networks on reproducible ray-traced corpora of 11 cities in two bands, with siting and material replicates, and has scored every model where a planner uses it. Tile labels of the log-distance law were shown to be ill-conditioned and, once their correlated errors are accounted for, imprecise: the cluster-robust standard error of the exponent is 3.2 times the naive one, and the exponent depends on the serving sites, the tile size and the cut-off of the data. Transferring tile parameters with learned regressors improves on the source median in 9 of 11 cities, although not significantly after correction for multiple comparisons; kernel transfer improves on it less, and the analysis of its limiting regimes and the gain from learning its metric locate its weakness in the isotropic distance. At the network level, even perfect tile parameters leave a path-loss error of 11.6 dB, assign only 50.1% of the locations to the correct serving site and misplace the cell edges by 76 m. A reachability hurdle removes most of the planning loss of the tile model and a line-of-sight dual-slope label part of the association error; the remainder is the geometry of each link. Site-aware hurdle link transfer, which predicts each link and whether it exists from its obstruction geometry, reduces the path-loss error to 9.3 dB, raises the serving-site accuracy to 68.4%, lowers the planning regret in every held-out city, and outperforms the per-link diffraction and 3GPP baselines built on the same geometry in path loss and serving site; a city-level certificate turns its accuracy into coverage that can be promised at a stated risk.

Future work will validate these findings on measured data with known site positions, extend the corpora to more cities, terrain, vegetation and antenna patterns, integrate the certified maps into the closed-loop planning and optimisation functions of a digital twin network, and study certificates that adapt to the target city from a small survey.

CRediT authorship contribution statement

Abraheem Rashid: Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Writing – original draft. **Faisal Iradat:** Software, Validation, Investigation. **Waseem Iqbal:** Supervision, Writing – review & editing. **Yawar Abbas Bangash:** Formal analysis, Writing – review & editing. **Bilgin Metin:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) to assist with software development, data analysis scripting and the drafting and editing of the manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability

The code, the scene definitions, the ray-traced path-loss corpora, the tile tables and every result file are available at `https://github.com/Abraheem13/terra-net`. Running `make reproduce` regenerates all numbers, tables and figures of this article and the article itself (Table 4 states the cost); `make corpus` regenerates the ray-traced corpora. Building footprints are from Overture Maps (release 2026-09-23.0), © OpenStreetMap contributors, Overture Maps Foundation, under the Open Database License.

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